

ESSENTIAL MATHEMATICS GRADE 10: PROBABILITY I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 3.0: Statistics and Probability
Sub-Strand	Sub-Strand 3.2: Probability I
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Equally likely events: performing experiments involving equally likely outcomes (e.g., tossing a coin, rolling a six-faced die) – Computing probability of equally likely outcomes – Range of probability: determining the range of probability of an event (0 to 1) – Mutually exclusive events: identifying and performing experiments involving mutually exclusive events in real-life situations – Probability of a single chance event involving mutually exclusive events – Independent events: performing experiments involving independent events (maximum of two experiments and two outcomes) – Probability of independent events in different situations – Application of probability in real-life situations including discussing the dangers of gambling
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - perform experiments involving equally likely outcomes in different situations, - compute probability of equally likely outcomes, - determine the range of probability of an event, - identify mutually exclusive events in real life situations, - perform experiments of single chance involving mutually exclusive events, - calculate the probability of an event involving a single chance, - perform experiments involving independent events, - determine the probability of independent events in different situations, - appreciate the role of probability in real-life situations.
Core Competencies	– Communication and Collaboration: the learner collaborates with peers when evaluating successes and failures from experiments involving mutually exclusive events to work out the probability of possible outcomes, and when sharing results from coin-toss and dice-roll experiments. – Critical Thinking and Problem Solving: the learner develops interpretation and inference skills when discussing the dangers of gambling using the knowledge of probability, and when determining the range of probability from real-life scenarios such as predicting rainfall in Kisumu or match outcomes for a Nairobi football team.
Core Values	– Responsibility: the learner makes correct and ethical decisions on life issues that involve gambling and chance, using the knowledge of probability to understand risk.

	– Unity: the learner collaborates respectfully with others when carrying out probability experiments such as tossing coins, rolling dice, and drawing cards, taking turns and recording results fairly.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners formulate questions about why outcomes of repeated chance experiments (e.g., tossing a coin 50 times) do not always split exactly 50/50, driving inquiry into theoretical vs. experimental probability. – Planning and Carrying Out Investigations: learners design and carry out coin-toss, dice-roll, and card-draw experiments, recording outcomes systematically in frequency tables to gather evidence about equally likely, mutually exclusive, and independent events. – Analysing and Interpreting Data: learners analyse experimental results from repeated trials, compare experimental probability to theoretical probability, and draw conclusions about the range and behaviour of probability values. – Engaging in Argument from Evidence: learners use probability data to construct arguments about the dangers of gambling, presenting evidence to peers that long-run odds favour the "house" in games of chance.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Decision Making): the learner makes informed decisions on real-life issues involving probability, such as understanding the dangers of gambling, assessing risk in sports betting, and evaluating chance in everyday situations common in Kenyan communities. – Financial Literacy: the learner applies the concept of probability to understand risk and expected value, particularly in the context of gambling and betting activities, reinforcing responsible financial behaviour.
Career Connections	– Insurance and Actuarial Science: probability is foundational to calculating risk, setting premiums, and underwriting policies — careers accessible through institutions such as the Actuarial Society of Kenya. – Sports Management and Coaching: probability underpins performance analysis, match prediction, and strategic decision-making in Kenyan sports such as athletics, football, and rugby. – Journalism and Media: journalists and data reporters use probability and statistics to report on polls, elections, health surveys, and sporting odds for Kenyan audiences. – Hospitality and Tourism Industry: probability concepts inform demand forecasting, overbooking strategies, and risk management in hotels and safari lodges across Kenya's tourism sector. – Guidance and Counselling: counsellors use probabilistic reasoning to communicate risk (e.g., health risk behaviours, gambling addiction) when advising students and community members.
Focus for Lessons	This unit introduces Grade 10 Essential Mathematics learners to the foundational concepts of probability through hands-on experiments, peer discussion, and real-life Kenyan contexts. Learners progress from understanding equally likely outcomes and the probability scale, through identifying and calculating probabilities for mutually exclusive events, to exploring independent events across two experiments with two outcomes. Throughout, the unit foregrounds responsible decision-making — particularly the mathematics behind why gambling is a losing proposition — anchoring abstract probability concepts in situations learners encounter in their communities.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: If past outcomes of a chance event — like a coin toss or a football match — do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling? KICD KEY INQUIRY QUESTION: 1. How do we apply probability in day-to-day life?
Anchoring Phenomenon	During the 2023 Kenyan Premier League season, sports commentators and fans frequently debated whether Gor Mahia FC's winning streak was "bound to end soon" — claiming the team was "due" for a loss after so many consecutive wins. At the same time, betting advertisements across Nairobi matatus and social media told youth that they could predict match outcomes and "beat

	the odds." Students are presented with this puzzling situation: if each football match is an independent event, why do so many people believe past results change the probability of future ones? And if a coin has landed on heads five times in a row, is tails now more likely? This seemingly simple question — do past outcomes change future chances? — is deeply counterintuitive and opens the door to exploring equally likely events, the range of probability, mutually exclusive events, and the mathematics of independence. The phenomenon is puzzling because everyday language and gambling culture actively promote the gambler's fallacy, yet probability mathematics tells a different story.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Equally Likely Events: Learners are introduced to the Gor Mahia / betting-odds phenomenon. They brainstorm what "chance" means in daily life (rain in Kisumu, drawing a card, tossing a coin). Working in pairs, learners toss a 10-shilling coin 40 times, record outcomes in a tally table, and calculate experimental probability of heads and tails. The class pools data and compares results to the theoretical probability of $\frac{1}{2}$. The Driving Question is posted on the Driving Question Board (DQB). Lesson 2 — Computing Probability of Equally Likely Outcomes: Learners roll a six-faced die 60 times in groups, record outcomes, and compute experimental probabilities for each face. They derive the formula $P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes}$. Worked examples use Kenyan contexts: drawing a name from a hat of 30 learners, picking a red ball from a bag of githeri-coloured tokens. Learners update the DQB with new understanding. Lesson 3 — Range of Probability: Learners investigate the boundaries of probability using real-life Kenyan scenarios — probability that the sun rises tomorrow over Mombasa (certain = 1), probability that it snows in Kisumu in July (impossible = 0), and intermediate cases. They construct a probability number line (0 to 1), place events on it, and write probability statements using fractions, decimals, and percentages. The idea that no experiment can yield $P > 1$ or $P < 0$ is formalised. DQB is updated. Lesson 4 — Mutually Exclusive Events (Identification & Concept): Learners examine the school raffle supporting phenomenon. Through discussion and sorting activities, they identify events that cannot happen simultaneously (mutually exclusive: rolling a 2 AND a 5 on one die; drawing a heart AND a spade in one draw) versus those that can. They list real-life Kenyan examples: a matatu cannot be in Nairobi and Mombasa at the same time; a learner cannot score both an A and a D on the same paper. The addition rule $P(A \text{ or } B) = P(A) + P(B)$ for mutually exclusive events is introduced with guided examples. Lesson 5 — Probability of Single-Chance Mutually Exclusive Events: Learners carry out experiments with a bag of coloured tokens (representing mandazi flavours: plain, coconut, cardamom) and a single draw. They calculate $P(\text{plain OR coconut})$ using the addition rule. Multiple worked examples are solved: a spinner divided into unequal sectors, a die showing an even OR a 5, selecting a day that is Monday OR Friday from a school week. Learners practise independently and peer-mark. DQB is updated with the addition rule. Lesson 6 — Independent Events (Concept & Experiments): The Kisumu rainfall supporting phenomenon is used to launch independent events. Learners carry out two-stage experiments: toss a coin AND roll a die; draw a card, replace it, draw again. They list all possible outcomes using tree diagrams and two-way tables (sample space). The rule $P(A \text{ and } B) = P(A) \times P(B)$ for independent events is derived from the data and formalised. Connections back to the DQ: why does a coin NOT "remember" its last result? Lesson 7 — Probability of Independent Events in Context: Learners apply the multiplication rule to two-experiment, two-outcome scenarios rooted in Kenyan life: probability that it rains on both Monday and Tuesday in Kisumu given daily $P(\text{rain}) = 0.7$; probability that two independently chosen Grade 10 learners are both female given school demographics; probability that a mandazi vendor sells out on two consecutive market days. Learners discuss the spinner/stall supporting phenomenon and

	calculate why the stall always profits in the long run. Dangers of gambling are discussed using probability evidence. DQB is substantially updated. Lesson 8 — Consolidation, Model Building & DQB Closure: Learners revisit the anchoring phenomenon (Gor Mahia / betting odds) and use all probability concepts learned to explain mathematically why each match is independent, why the gambler's fallacy is false, and why gambling is designed to be a losing proposition. Groups present mini-posters connecting equally likely events, range of probability, mutually exclusive events, and independent events to the Driving Question. The DQB is reviewed and closed. A class written assessment covers all SLOs; learners also complete a short reflection on one real-life decision they would make differently using probability knowledge.
Supporting Phenomena	– A mandazi vendor in Kericho market tells a customer: "I've flipped this 10-shilling coin six times and it landed tails every time — heads must come next!" Students investigate whether this claim is mathematically valid, using repeated coin-toss experiments to explore equally likely outcomes and the range of probability. – During a class raffle for a school trip to Lake Victoria, two learners wonder: "If my name is drawn for Prize 1, can it also win Prize 2?" This supports exploration of mutually exclusive versus non-mutually exclusive events and single-chance probability. – A student notices that the probability of rain on any given day in Kisumu during April (historically ~70%) does not change based on whether it rained the day before. This real Kenyan weather context supports understanding of independent events and how probabilities are combined across two trials. – A Nairobi school's sports day features a spinner game at the fundraising stall — spin to win a free soda or mandazi. Students analyse the spinner's sections to determine whether the game is fair, connecting equally likely outcomes, range of probability, and the idea that the stall (like a casino) is always designed to win in the long run.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 × 40-minute lessons)): Anchoring Phenomenon & Equally Likely Events: Is Gor Mahia "Due" for a Loss?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To anchor the unit in the Gor Mahia / sports-betting phenomenon and establish the foundational concept that equally likely events have a fixed theoretical probability regardless of past outcomes.
Knowledge	Learners will know that: (a) equally likely outcomes each have the same chance of occurring; (b) theoretical probability of an equally likely event = (number of favourable outcomes) ÷ (total number of possible outcomes); (c) experimental probability is calculated from trials and may differ from theoretical probability; (d) as the number of trials increases, experimental probability approaches theoretical probability (Law of Large Numbers, introduced informally).
Skills	Learners will be able to: (a) perform a coin-toss experiment, record outcomes in a tally table, and calculate experimental probability of heads and tails; (b) compare experimental probability to the theoretical probability of $\frac{1}{2}$; (c) pool class data and observe the effect of increasing sample size; (d) articulate in writing why past outcomes of an independent event do NOT change its future probability.
Attitudes	Learners will appreciate that mathematics provides an objective tool for evaluating claims about chance, and will begin to develop a critical, evidence-based stance towards gambling advertisements and the gambler's fallacy. (Value: Responsibility — making correct decisions on life issues that involve gambling using the knowledge of probability.)
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	This is the opening lesson of the unit. It anchors all subsequent learning in the real-world Gor Mahia / Nairobi-betting phenomenon. Learners encounter the central puzzle — do past outcomes change future chances? — before any formal instruction, ensuring genuine curiosity. The coin-toss experiment generates personal data that learners will return to throughout the unit as they build increasingly sophisticated probability models. The Driving Question Board (DQB) opened today becomes the living record of the class's growing understanding across all lessons.
Safety Notes	No chemical or physical hazards. Remind learners to handle 10-shilling coins hygienically (wash hands after handling coins). Ensure learners toss coins onto the desk surface — not across the room — to avoid distractions or injury.

B. LESSON OVERVIEW

In this opening lesson, learners encounter the anchoring phenomenon: during the 2023 Kenyan Premier League season, fans claimed Gor Mahia FC was "due" for a loss after a winning streak, while betting advertisements on Nairobi matatus and social media promised youth they could "beat the odds." Learners first predict whether past results of a coin toss can change the probability of future outcomes. They then conduct a structured coin-toss experiment (40 tosses per pair, using a 10-shilling coin), record outcomes in a tally table, and calculate experimental probability. Class data are pooled to compare experimental results with the theoretical probability of $\frac{1}{2}$. The lesson closes with the formal opening of the Driving Question Board (DQB) and an introductory model sketch connecting the phenomenon to the mathematics of equally likely events. No prior probability knowledge is assumed; this lesson surfaces and challenges everyday misconceptions before formal teaching begins.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Learners read a short printed or on-screen scenario card (2–3 sentences) describing two situations: (A) Gor Mahia FC has won their last 7 Premier League matches. A fan says, 'They are definitely due for a loss next week — you can't keep winning forever!' (B) A matatu in Nairobi displays a betting-app advert: 'Use our stats — predict the next winner and earn KSh 5,000 today!' Working individually (3 minutes, silent), learners write answers to two prompts in their exercise books: (1) 'Do you agree with the fan? Is Gor Mahia now MORE likely to lose? Explain your thinking.' (2) 'If you toss a 10-shilling coin and it lands on heads five times in a row, is tails now more likely on the sixth toss? Why or why not?' Learners then share with a partner (Think-Pair-Share, 2 minutes). Three to four pairs are cold-called to share their partner's prediction with the class.	1. Display or distribute the scenario card. Read it aloud with expression, pausing at the fan's quote. Ask: 'Has anyone seen adverts like this on a matatu or on their phone? What do they promise?' (Allow 2–3 hands, 30 seconds.) 2. Issue the two written prompts. Say exactly: 'Work alone — no talking, no looking at your neighbour's book. I want YOUR honest thinking before we do any maths. You have 3 minutes.' Start a visible timer. 3. After 3 minutes: 'Now share your answer to question 1 with the person next to you. Listen carefully — you will be asked to report your PARTNER'S idea, not your own. 2 minutes.' 4. After pair-share, cold-call exactly 4 pairs using the class register (do not ask for volunteers): 'Amina, tell me what YOUR PARTNER said about Gor Mahia being due for a loss.' WAIT 10 seconds after each name before prompting. Record key student ideas (verbatim) on the board under two columns: 'Past results DO change future chances' vs 'Past results DON'T change future chances.' 5. Do NOT correct misconceptions yet. Say: 'We are going to use mathematics today to find out who is right. Keep your predictions — we will return to them at the end.'	Think-Pair-Share with cold-call reporting. This strategy surfaces prior knowledge and misconceptions (especially the gambler's fallacy) before instruction, giving the teacher diagnostic information and creating cognitive dissonance that motivates the investigation. Recording student ideas on the board validates all voices and sets up a public record to revisit.	OBSERVABLE INDICATOR: Teacher scans written responses during the 3-minute silent period and notes (on a clipboard or sticky note) how many learners in each row wrote that tails becomes 'more likely' after five heads — this is the key misconception to track. Cold-call responses reveal the depth of reasoning. TARGET: All learners have committed a written prediction to their books before the experiment begins.
Observe Phase  VIDEO: Die rolling probability with independent events Source: Khan Academy (English - US curriculum)	Working in pairs, learners conduct a structured coin-toss experiment. Each pair receives one 10-shilling coin. Learner A tosses the coin; Learner B records the outcome (H or T) using tally marks in a pre-drawn table in their exercise books. They complete 40 tosses, then swap roles to verify. After	1. Before distributing coins, say: 'We are going to gather EVIDENCE — just like a scientist at the Kenya Medical Research Institute gathers data before drawing conclusions. Your job is to record every single outcome honestly. No cheating the toss.' 2. Draw the tally table on the board:	Pooled Data Aggregation. By combining all pairs' data into a single class table, learners directly observe the Law of Large Numbers informally: as the number of trials increases, experimental probability approaches 0.50. This makes abstract theory tangible and	OBSERVABLE INDICATOR: Each pair's tally table is complete (40 entries, no gaps) and their calculated experimental probability is a decimal between 0 and 1. Teacher checks 6–8 pairs' tables during circulation — look specifically for: (a) correct tally grouping in fives, (b) correct

Search ARES for similar videos EXERCISE: Theoretical and Experimental Probability Practice Source: CK-12 Search ARES for similar readings	recording, each pair calculates: (i) Number of Heads, (ii) Number of Tails, (iii) Experimental P(Heads) = (Number of Heads) ÷ 40, (iv) Experimental P(Tails) = (Number of Tails) ÷ 40. Results are written on a sticky note or called out to the class recorder, who fills in a shared class tally table on the board (one row per pair). The class then calculates the pooled totals and pooled experimental probabilities using all pairs' data combined.	columns = Toss Number (groups of 5), Outcome (H/T), Running Total H, Running Total T. Say: 'Copy this table NOW. Leave space for 40 rows.' Allow 2 minutes for table setup. 3. Distribute coins. Say: 'Hold the coin at the edge, flip it with your thumb, catch it on the back of your other hand. Do NOT spin it on the desk — that changes the physics.' Demonstrate once. 4. Circulate during the experiment. Every 3–4 minutes, narrate: 'Pair at the back — I can see you have 15 heads out of 20. What is your experimental probability of heads so far? [WAIT 10 seconds.] Does that surprise you?' Ask at least 6 different pairs a similar running-probability question during the activity. 5. After 40 tosses per pair: 'Stop. Calculate your experimental P(Heads) and P(Tails) now. Round to 2 decimal places.' Allow 3 minutes. 6. Call out results pair by pair and fill the class table on the board. Ask the class recorder (a learner) to compute the column totals aloud. Then ask: 'What is our CLASS experimental probability of heads?' Cold-call one learner to compute it on the board. WAIT 15 seconds. 7. Write the theoretical probability: 'Mathematics tells us $P(\text{Heads}) = 1/2 = 0.50$. How does our class result compare?' Pause for reactions.	Kenya-grounded (their own coin, their own tosses). It also mirrors how actuaries and insurance companies in Nairobi use large datasets to estimate risk.	division (e.g., $22 \div 40 = 0.55$, not $22/40$ left as a fraction without simplification). RED FLAG: Any pair whose total $H + T \neq 40$ — stop and correct immediately before they submit to the class table.
Explain Phase VIDEO: Experimental probability Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners return to their original predictions from Phase 1. The teacher facilitates a whole-class discussion connecting the coin-toss data to the Gor Mahia phenomenon. Learners are guided to: (a) define 'equally likely outcomes' using their coin-toss	1. Point to the two columns on the board from Phase 1. Say: 'Look at what you predicted before the experiment. Now look at our data. What does the evidence tell us?' Cold-call 3 learners — one from each prediction column and one who left a 'not sure' response.	Claim-Evidence-Reasoning (CER) Writing Frame. Learners construct an explanation paragraph using the structure: Claim (the fan is wrong) → Evidence (our coin-toss data, theoretical formula) → Reasoning (each toss is independent; past outcomes don't	OBSERVABLE INDICATOR: The explanation paragraph must contain three elements to be assessed as 'Meets Expectations': (1) a correct statement of the probability formula, (2) use of the word 'independent' or 'does not change,' (3) a direct reference

 HTML: Probability of Independent Events (Flexbook) Source: CK-12  Search ARES for similar readings	experience; (b) state the formula for theoretical probability of equally likely events: $P(\text{event}) = (\text{number of favourable outcomes}) \div (\text{total number of outcomes})$; (c) apply the formula to: a coin toss ($P(H) = 1/2$), rolling a fair die ($P(3) = 1/6$ — introduced as an extension example), and the Gor Mahia match scenario (is a football match an equally likely event?). Learners write a structured 'Explanation Paragraph' in their exercise books (individual, 5 minutes): 'The fan said Gor Mahia was due for a loss. Using what we learned today, I can explain that...'	WAIT 12 seconds each time. 2. Ask: 'If we tossed this coin another 40 times, would it REMEMBER the results from the first 40 tosses?' WAIT 10 seconds. Take 2 responses. Then say firmly: 'A coin has no memory. Each toss is a fresh, independent event. The mathematics does not change.' Write on the board: 'Each toss is INDEPENDENT. Past outcomes do NOT change future probability.' 3. Introduce the formula. Say: 'In mathematics we define: $P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes}$. For our coin: $P(\text{Heads}) = 1 \div 2 = 0.5$. Always. After 5 heads in a row — still 0.5.' Write this on the board with a box around it. 4. Ask: 'Now think about Gor Mahia. Is a football match the same as a coin toss? Are all outcomes equally likely?' WAIT 15 seconds. Take 3 responses. Guide learners to notice that football matches involve skill, fitness, home advantage — they are NOT simple equally likely events. Say: 'We will explore this deeper in later lessons. For now, write your explanation paragraph.' 5. After 5 minutes of individual writing, cold-call 2 learners to read their paragraph. Offer specific praise: 'Excellent — Fatuma used the word INDEPENDENT correctly.' Correct any learner who writes 'tails was more likely' with a direct, calm restatement: 'Evidence shows $P(\text{Heads})$ stayed close to 0.5 throughout. The coin does not know what it did before.'	change P). This NGSS Science and Engineering Practice (Constructing Explanations) bridges the concrete experiment to the abstract mathematical principle and directly dismantles the gambler's fallacy.	back to the coin experiment or Gor Mahia scenario. Teacher reads 4–6 paragraphs as learners write (circulate and read over shoulders). Mark a small tick on the margin if all three elements are present; write a question mark if the gambler's fallacy is still embedded in the writing.
Driving Question Board (DQB) Creation	The teacher formally introduces the unit Driving Question: 'If past outcomes of a chance event — like a coin toss or a football match	1. Read the Driving Question aloud slowly. Ask: 'Which part of this question does today's experiment begin to answer?'	Driving Question Board (DQB) Launch. The DQB is a public sensemaking artefact that externalises the class's collective	OBSERVABLE INDICATOR: Every learner has written at least two genuine questions in their exercise book (teacher checks

 VIDEO: Random numbers for experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probability of Independent Events Practice Source: CK-12  Search ARES for similar readings	— do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling?' The question is displayed prominently (on a poster, whiteboard section, or projected). Learners individually write TWO questions of their own in their exercise books — things they are still wondering about after today's lesson. Selected questions are written on sticky notes (or paper strips) and posted on the class DQB. The DQB is organised into three columns: 'What We Wonder,' 'What We Are Learning,' 'What We Now Know.' Today, all questions go into 'What We Wonder.'	WAIT 12 seconds. Take 2 responses. Say: 'Today we answered the first part — past outcomes do NOT change future probability. But we have not yet answered: how do we predict? And how does this protect us from gambling? Those answers will come in the lessons ahead.' 2. Say: 'I want every person to write TWO questions in your book right now. Things you are genuinely curious about after today. You have 2 minutes. They can be about football, betting, dice, cards — anything related to chance.' Start timer. 3. After 2 minutes, ask pairs to share their best question with each other, then select one to write on a paper strip. Collect strips and pin/tape them to the DQB under 'What We Wonder.' Read 5–6 questions aloud to the class. 4. Say: 'Every time we learn something new in this unit, we will move a question from WONDER to LEARNING to KNOW. By the end of this sub-strand, this board will show the journey your thinking has taken.' 5. Take a photo of the DQB (teacher's phone) for documentation. Tell learners: 'This board stays up for the whole unit. Add to it any time you have a new question — even outside class.'	curiosity and tracks growing understanding across the unit. Placing all questions in 'What We Wonder' today creates productive uncertainty — learners know there is more to discover. The physical board (or digital equivalent) makes the learning journey visible and motivates persistence.	books during the 2-minute writing period). QUALITY INDICATOR: Questions that move beyond 'What is probability?' to questions such as 'Why do betting companies always win in the end?' or 'What happens if you toss a coin 1,000 times?' indicate higher-order thinking. Teacher notes these learners for extension tasks in subsequent lessons.
Model Building Phase  VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners create their first individual model (a labelled diagram or annotated sketch) in their exercise books. The model must represent: (a) a coin toss as a system with two equally likely outcomes; (b) the probability value for each outcome ($P = 1/2$); (c) an arrow or annotation showing that 'past outcomes have NO influence on the next toss' (independence); (d) a connection back to the Gor	1. Say: 'Scientists and mathematicians use MODELS to show how a system works. Your model today does not need to be beautiful — it needs to be ACCURATE. Draw a coin. Show the two outcomes. Label the probability of each. Then add an arrow showing what the coin does NOT know.' 2. Demonstrate a partial model on the board (deliberately leave out the	Progressive Model Revision (NGSS Science and Engineering Practice: Developing and Using Models). By building an explicit, labelled model on Day 1, learners create a cognitive anchor. The instruction to revisit and revise the model in each subsequent lesson makes conceptual change visible — learners can literally see their own understanding grow. Peer review introduces scientific	OBSERVABLE INDICATOR: Model Version 1 must contain four elements for 'Meets Expectations': (1) a coin or circle split into two equally labelled sections, (2) $P = 1/2$ (or 0.5) written for at least one outcome, (3) any annotation conveying 'no memory' / 'independent,' (4) any written or drawn reference to the Gor Mahia/football scenario. Teacher collects exercise books for 5–10

 HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Mahia scenario — learners draw or write how the coin-toss logic relates (or does NOT relate) to predicting a football match. Models are shared with a partner for a 1-minute peer-review ('What did your partner include that you missed?'), then a revised version is starred in their book as 'Model Version 1.'	independence arrow). Say: 'What is missing from my model?' WAIT 10 seconds. Cold-call one learner. Add the independence annotation based on their suggestion, saying: 'Exactly — I need to show that the coin has NO MEMORY.' 3. Allow 5 minutes for individual model drawing. Circulate and offer targeted prompts: 'Can you show me WHERE in your model the number 0.5 comes from?' / 'How does your model show that tossing heads five times in a row does NOT make tails more likely?' 4. After 5 minutes, run 1-minute peer review. Say: 'Look at your partner's model. Tell them ONE thing they included that is correct and ONE thing they should add or fix. You have 1 minute — go.' 5. Say: 'Put a star next to your model and write: Model Version 1 — [today's date]. We will update this model in every lesson as we learn more. By Lesson 5, your model will show the full picture of probability — including mutually exclusive and independent events.' 6. Close the lesson: 'So — is Gor Mahia due for a loss? What does the mathematics say?' Cold-call one learner for the final word. WAIT 15 seconds. Affirm the correct answer and connect to the DQB: 'That question is now on our board. Let us keep investigating.'	discourse norms (evidence-based critique) in a low-stakes format.	learners at random at the end of the lesson to review models before the next class. Findings are used to plan targeted questioning in Lesson 2.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and write brief notes in your planning diary: (1) MISCONCEPTION TRACKING — How many learners still held the gambler's fallacy in their explanation paragraph (Phase 3)? What specific language did they use? Plan a direct re-address at the opening of Lesson 2 using one of those anonymised examples. (2) DATA QUALITY — Did the pooled class experimental probability fall between 0.45 and 0.55? If it fell outside this range, consider whether coin-tossing technique was inconsistent (spinning rather than flipping). Model correct technique again in Lesson 2. (3) DQB RICHNESS — Were the learners' questions mostly surface-level ('What is probability?') or did any go deeper ('Why do betting companies make money if every event is 50-50?')? The richer questions signal learners who are ready for extension into independent events. (4) MODEL COMPLETENESS — Of the 5–10 books reviewed, how many models included all four

required elements? If fewer than 60% included the independence annotation, open Lesson 2 with a whole-class model gallery walk. (5) ENGAGEMENT AND INCLUSION — Were all learners actively tossing and recording, or did some pairs have one dominant partner? In Lesson 2, assign roles explicitly (Tosser, Recorder, Calculator, Reporter) to ensure equitable participation. (6) PCI CONNECTION — Did the gambling/betting discussion remain educational and non-stigmatising? If any learner disclosed personal family experience with gambling, follow up privately and liaise with the school counsellor in line with Life Skills PCI guidance.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners tossed a 10-shilling coin 40 times per pair, recorded outcomes in a tally table, and pooled class results to compare experimental probability with the theoretical value of $P(\text{Heads}) = \frac{1}{2}$.
What did I learn?	Each coin toss is an independent equally likely event. Theoretical probability = (favourable outcomes) $\div$ (total equally likely outcomes). Past outcomes do NOT change future probability — a coin has no memory — directly contradicting the gambler's fallacy promoted in betting advertisements seen on Nairobi matatus and social media.
How does this explain the phenomenon?	The Gor Mahia fan's claim that the team is 'due' for a loss is an example of the gambler's fallacy. Mathematics shows that if each event is independent and equally likely, its probability remains fixed regardless of previous outcomes. Our coin-toss data supported the theoretical $P = 0.5$, especially when the whole class's results were pooled — showing that larger samples give more reliable estimates of true probability.

LESSON 2 (40 minutes): Computing Probability of Equally Likely Outcomes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners derive and apply the formula $P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes}$, using hands-on die-rolling experiments and Kenyan contexts, building the mathematical foundation needed to challenge the gambler's fallacy introduced in the anchoring phenomenon.
Knowledge	Learners know that equally likely outcomes are outcomes that have the same chance of occurring; that experimental probability is computed from recorded trials; and that theoretical probability is derived from the formula $P(\text{event}) = \text{favourable outcomes} \div \text{total equally likely outcomes}$.
Skills	Learners can perform a die-rolling experiment, record and tally outcomes systematically, compute experimental probabilities as fractions, decimals and percentages, and apply the theoretical probability formula to Kenyan real-life contexts such as drawing a name from a class register or picking a coloured token from a bag.
Attitudes	Learners appreciate that mathematics — not intuition, superstition, or betting advertisements — gives us reliable tools for predicting chance events, and develop responsibility in making evidence-based decisions about risk.
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	In Lesson 1, learners explored the anchoring phenomenon — the gambler's fallacy in Gor Mahia FC's winning streak and Nairobi betting culture — and placed their initial questions on the Driving Question Board (DQB). Lesson 2 now gives learners the mathematical tool to begin answering those questions: the probability formula for equally likely outcomes. By rolling a die 60 times and comparing experimental results to theoretical predictions, learners discover that each face always has a 1-in-6 chance regardless of what happened before — directly challenging the 'due for a loss' belief and 'beat the odds' gambling claims from the phenomenon.
Safety Notes	Ensure dice and tokens do not become projectiles; remind learners to roll gently on the desk surface. No sharp objects are used. Githeri-coloured tokens (pebbles, dried beans, or paper squares) are choking hazards — remind learners not to place them in their mouths.

B. LESSON OVERVIEW

Learners work in groups of five to roll a six-faced die 60 times, tally outcomes, and compute experimental probabilities for each face. Through guided discussion, they derive the theoretical probability formula $P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes}$. The teacher then anchors the formula in two Kenyan worked examples: (1) drawing a name from a hat containing 30 learners' names, and (2) picking a coloured token from a bag of githeri-coloured tokens (representing the mixed colours of a githeri serving — white maize, brown beans, green peas). Learners connect their findings back to Gor Mahia FC's winning streak and the betting advertisements, updating the Driving Question Board with new mathematical reasoning. The lesson advances the 8-lesson evidence-gathering sequence by equipping learners with the core formula they will use in all subsequent probability lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Each group receives a standard	Distribute one die per group. Say:	Prediction Anchoring — by	Observable indicator: Every

 VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	six-faced die. Before rolling, learners individually write predictions in their exercise books: (a) If you roll this die 60 times, how many times do you predict each face (1–6) will appear? Record your predicted count for each face. (b) If Gor Mahia FC has won their last 7 matches, does the probability of winning the next match change? Write YES or NO and a one-sentence reason. Learners share predictions with their group partner before the class shares out.	'Before we roll a single time, I want each of you — working alone, in silence — to predict how many times each face will appear if we roll 60 times. Write your six predictions in a table in your book. You have 90 seconds.' [Allow 90 seconds of silent individual work.] Then say: 'Now turn to your neighbour and compare predictions. Do you agree? Disagree? Why?' [Allow 60 seconds of pair talk.] Cold-call 3 learners using name sticks: 'Akinyi, what did you predict for face number 6?' — 'Omondi, did you predict the same number for every face, or different numbers? Why?' — 'Fatuma, how did you decide on your number?' [WAIT TIME: 12 seconds after each question before accepting an answer.] Record 3–4 different class predictions on the board without evaluating them. Then say: 'Keep your predictions — at the end of this experiment, we will come back and see who was closest, and more importantly, WHY.'	committing individual written predictions before the experiment, learners activate prior intuition (often reflecting the gambler's fallacy or vague symmetry thinking). This creates cognitive dissonance when experimental data diverges from prediction, making the theoretical formula more memorable and meaningful.	learner has a completed prediction table (6 faces, 6 predicted counts) in their exercise book before rolling begins. Teacher scans the room to confirm all books show written predictions. Listen for prediction reasoning: learners who say 'each face should get 10 because there are 6 faces and 60 rolls' are already reasoning proportionally — note these learners for targeted questioning in the Explain Phase.
Observe Phase  VIDEO: Intro to theoretical probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Theoretical and Experimental Probability Practice Source: CK-12  Search ARES	Groups of five roll the die 60 times, with roles assigned: Roller (rolls the die), Caller (calls out the number aloud), Tally Keeper (marks tallies on the group recording sheet), Checker (confirms the tally keeper's mark each time), and Timer/Counter (counts rolls aloud every 10 rolls). After 60 rolls, each learner independently computes the experimental probability for each face as a fraction (e.g., 11/60), a decimal (rounded to 2 d.p.), and a percentage. Groups then compare their experimental probabilities with the teacher-posted class aggregate (teacher compiles all	Say: 'Your group has a die and a recording sheet. I am going to assign each person a role — listen carefully.' [Assign roles by pointing to each position in the group.] 'The Roller rolls gently on the desk — not on the floor, not in the air. The Caller reads the number loudly so the whole group hears. The Tally Keeper marks the tally. The Checker confirms every single mark. The Timer counts every 10 rolls and announces it. Ready? You have 8 minutes. Begin.' [Circulate to all groups. At the 4-minute mark, announce: 'You should be at approximately 30 rolls by now — check your count.'] After	Data Aggregation and Pattern Recognition — individual group data (60 rolls) is pooled into a class aggregate (potentially 300–360 rolls across 5–6 groups). This strategy helps learners see that larger sample sizes produce experimental probabilities that converge toward the theoretical value of $1/6 \approx 0.167$, directly modelling the Law of Large Numbers without naming it formally, and setting up the derivation of the theoretical formula.	Observable indicators: (1) Each learner's book contains a completed tally table and three rows of probability expressions (fraction, decimal, percentage) for all six faces. (2) During circulation, check that fractions are written with denominator 60 (or the group's total rolls), not simplified incorrectly. (3) Listen for learners who say 'all the fractions are close to 10/60' — these learners are ready for the formula derivation. Red flag: learners writing probabilities greater than 1 or expressing probability as a count rather than a ratio — address immediately with: 'Probability is

for similar readings	groups' tallies on the board to create a larger combined sample). Learners also observe a short comparison: a bag of githeri-coloured tokens (10 white tokens = maize, 8 brown tokens = beans, 2 green tokens = peas, totalling 20) is passed to each group. Learners predict, then draw one token blindly, record its colour, replace it, and repeat 20 times, recording outcomes.	rolling: 'Stop. Now, working individually — not copying your neighbour — compute the experimental probability for each face. Express it as a fraction over 60, then as a decimal to 2 decimal places, then as a percentage. You have 3 minutes.' [Collect one tally from each group. Write the combined class tallies for each face on the board.] Cold-call 2 learners: 'Chebet, what experimental probability did your group get for face number 3? Express it as a fraction.' [WAIT TIME: 10 seconds.] 'Mutua, convert that fraction to a decimal for us.' [WAIT TIME: 12 seconds.] Then introduce the token bag: 'This bag contains tokens in three colours — like the colours in a good githeri: white for maize, brown for beans, green for peas. Before you draw, count the tokens I pour out so you know exactly how many of each colour there are.' [Pour tokens onto a tray, let groups count.] 'Now do 20 draws with replacement. Record each outcome.'		always a part of a whole — what is the whole here?
Explain Phase  VIDEO: Die rolling probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Using the class aggregate data on the board, learners are guided to articulate the pattern: each face's experimental probability is approximately 1/6. The teacher then introduces the theoretical probability formula: $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of equally likely outcomes}}$. Learners apply the formula to two Kenyan worked examples in their books: (1) Worked Example A — 'Ms Wanjiku writes all 30 learners' names on separate slips of paper and places them in a hat to select one learner to carry the class register to the	Point to the class aggregate on the board. Say: 'Look at our combined class data. What do you notice about each face's experimental probability? Discuss with your partner for 30 seconds, then I will ask.' [30 seconds pair talk. Cold-call 2 learners: 'Otieno, what pattern do you see?' WAIT TIME: 12 seconds. 'Njeri, do you agree or disagree with Otieno, and why?' WAIT TIME: 12 seconds.] Then say: 'You have discovered something important. Each face has 1 chance out of 6 equally likely outcomes. That gives us a formula.' [Write on the board in	Formula Derivation from Data (Inductive Reasoning) — rather than presenting the formula as a rule to memorise, the teacher guides learners to extract it from their own experimental data. This inductive path (data → pattern → formula) mirrors NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data) and Practice 6 (Constructing Explanations), ensuring the formula has conceptual roots rather than being an abstract prescription. The Kenyan worked examples (class register, githeri tokens) further ground the formula	Observable indicators: (1) Formula is correctly written and underlined in every learner's book. (2) Worked Example A answer is $P = 1/30$, correctly labelled. (3) Worked Example B answers are $P(\text{white}) = 10/20 = 1/2$, $P(\text{brown}) = 8/20 = 2/5$, $P(\text{green}) = 2/20 = 1/10$, and learners confirm the sum equals 1 ($10/20 + 8/20 + 2/20 = 20/20 = 1$). (4) The Gor Mahia sentence explicitly references 'equally likely outcomes' and the formula structure. Red flag: learners writing $P = 30/1$ (inverted fraction) — address by re-asking 'Which is the part and which is the whole?'

	staffroom. What is the probability that your name is selected?' (2) Worked Example B — 'A bag contains 20 githeri-coloured tokens: 10 white (maize), 8 brown (beans), 2 green (peas). One token is drawn at random. What is the probability of drawing (a) a white token, (b) a brown token, (c) a green token? Do all three probabilities sum to 1? Why?' Learners then return to the Gor Mahia FC phenomenon: 'Using what you now know, explain in one mathematical sentence why Gor Mahia's previous 7 wins do NOT change the probability of the next match result.'	large print:] $P(\text{event}) = \text{Number of favourable outcomes} \div \text{Total number of equally likely outcomes}$. Say: 'Copy this formula into your books and underline it. This is the mathematical engine of probability.' [Pause 30 seconds for copying.] 'Now open to a fresh page. We will do two worked examples together.' [Read Worked Example A aloud. Ask: 'How many favourable outcomes are there for YOUR name being drawn? How many total equally likely outcomes?' WAIT TIME: 10 seconds. Cold-call 1 learner: 'Kamau, give me the fraction.' Confirm: $P = 1/30$.] [For Worked Example B, say: 'Work this one out individually — parts (a), (b), and (c). Then check whether your three answers sum to 1. You have 3 minutes.' After 3 minutes, cold-call 3 different learners for parts (a), (b), (c). WAIT TIME: 10 seconds each. Then ask: 'Amina, do all three probabilities sum to 1? What does that tell us?' WAIT TIME: 12 seconds.] Finally say: 'Return to Gor Mahia FC. In your book, write ONE mathematical sentence connecting this formula to why their past wins do not change the next match's probability. Use the word equally likely in your sentence. You have 90 seconds.'	in familiar, everyday contexts learners trust.	
Driving Question Board (DQB) Creation  VIDEO: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Each learner writes two sticky-note contributions (or index card entries if sticky notes are unavailable): (1) a CLAIM card — one thing they now know for certain about probability, expressed as a mathematical statement using today's formula; and (2) a QUESTION card — one new question today's lesson raised for them. Groups share their cards,	Say: 'Take two cards — one yellow, one blue if available, otherwise just two separate pieces of paper. On the first card, write a CLAIM — something you are now mathematically confident about. Start your sentence with: $P(\text{event}) = \dots$ On the second card, write a QUESTION — something today raised that you still wonder about. Connect your question to Gor	Driving Question Board Curation — the DQB functions as a public, cumulative record of the class's growing understanding. Physically moving Lesson 1 cards to 'Solved' gives learners a visible, tangible sense of intellectual progress. New question cards honour learners' emerging curiosity (e.g., 'What if outcomes are NOT equally likely?' — previewing mutually exclusive	Observable indicators: (1) Every learner produces at least one CLAIM card that correctly uses the formula $P(\text{event}) = \text{favourable outcomes} \div \text{total equally likely outcomes}$ with a specific example. (2) QUESTION cards reference the phenomenon (Gor Mahia, coin toss, or betting) rather than being purely abstract. (3) Teacher checks that no claim card contains

 HTML: Use Simulations to Explore Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	then class representatives pin or tape them to the DQB under two columns: 'What We Now Know' and 'What We Still Wonder'. The teacher draws learners' attention to any DQB questions from Lesson 1 that today's formula has now answered, and physically moves those cards to a 'Solved' section.	Mahia, the coin toss, or the betting adverts if you can. You have 2 minutes of silent individual writing.' [After 2 minutes: 'Read your claim to your partner. Does your partner agree it is mathematically correct? If not, fix it together.'] [After 60 seconds: 'Bring your cards. Let us update the board.'] [As learners pin cards, read 3 claim cards aloud and confirm or gently correct them. Read 2 question cards aloud and say: 'This is an excellent question — hold onto it. We will have the mathematics to answer it in Lesson 3 or 4.'] Point to Lesson 1 DQB questions about 'what does equally likely mean?' and say: 'Jamil asked in Lesson 1 whether all outcomes must be equally likely for probability to work. Can anyone use today's formula to answer that?' [WAIT TIME: 12 seconds. Cold-call 1 learner.] Move answered cards to the 'Solved' column visibly.	and independent events in later lessons) and demonstrate that science and mathematics advance through questions, not just answers.	a probability value greater than 1 or less than 0 — if found, address with the whole class as a teachable moment before the end of the phase.
Model Building Phase  VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the probability model they began sketching in Lesson 1 (a simple diagram or written description of what determines the outcome of a chance event). They now add to this model: (1) They annotate their diagram with the formula $P(\text{event}) = \text{favourable outcomes} \div \text{total equally likely outcomes}$, placing it explicitly on the model. (2) They add a new component — a labelled example showing a sample space (all possible outcomes of rolling a die: {1, 2, 3, 4, 5, 6}) with the favourable outcomes circled. (3) They write one sentence connecting their model to the anchoring phenomenon: 'Each Gor Mahia match result belongs to a sample	Say: 'Get out the model you started in Lesson 1. If you do not have it, redraw the basic structure now — you have 60 seconds.' [Circulate to confirm all learners have a starting model.] 'Now you are going to upgrade your model with three additions. Listen to each one before you draw.' [State addition 1. Pause 45 seconds for learners to annotate.] [State addition 2 — sample space with die faces. Pause 60 seconds.] [State addition 3 — the Gor Mahia sentence. Say: 'This sentence is the heart of what we are building toward. Take your time — 90 seconds.' Pause.] After model work: 'Share your model with your partner. Your partner will tell you one strength and one improvement	Cumulative Model Revision (NGSS Practice 2: Developing and Using Models) — the model is not redrawn from scratch each lesson but iteratively annotated and extended. This mirrors how scientific models evolve with new evidence. Requiring learners to explicitly write the Gor Mahia sentence on their model ensures the abstract formula stays tethered to the anchoring phenomenon and does not become a decontextualised procedure.	Observable indicators: (1) Model includes the probability formula correctly stated and labelled. (2) Sample space for a die is listed as {1, 2, 3, 4, 5, 6} (or equivalent notation) with at least one favourable outcome circled and its probability computed. (3) The Gor Mahia sentence explicitly uses the words 'sample space', 'equally likely', and 'formula' (or 'P(event) ='). (4) Peer feedback is specific (not just 'good job') — teacher circulates and prompts vague feedback: 'Can you point to something specific on the model that shows that?'

	space of equally likely outcomes; the formula tells us the probability of any single result, and past results do not change the total number of equally likely outcomes in the NEXT match.' Learners share their updated model with a partner and give one piece of feedback using the sentence stem: 'Your model shows _____. One thing to add or change is _____.'	using the sentence stem on the board.' [Write stem on board: 'Your model shows _____. One thing to add or change is _____.'] [Allow 90 seconds of paired feedback.] Cold-call 2 learners to share their partner's feedback: 'Baraka, what did your partner suggest you add to your model?' [WAIT TIME: 10 seconds.] Close by saying: 'Your models are growing. Each lesson in this sequence, your model will become more powerful — until by Lesson 8 it can explain everything the betting companies and the football commentators get wrong.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) When learners computed experimental probabilities from their die rolls, did the experimental values cluster near 1/6, or were there significant deviations? What does this tell you about the importance of sample size — and how will you address the Law of Large Numbers idea in a future lesson without overwhelming learners now? (2) Did any learner's Gor Mahia sentence on their model reveal persistent gambler's fallacy thinking (e.g., 'past wins make the sample space smaller')? If so, plan a targeted one-on-one conversation before Lesson 3. (3) Were the githeri-coloured token probabilities (1/2, 2/5, 1/10) within reach of all learners, or did some struggle with fractions with different denominators? If fraction fluency is a barrier, consider a brief 5-minute fraction review warm-up at the start of Lesson 3. (4) Review the DQB QUESTION cards: do the learners' new questions naturally lead toward mutually exclusive events (Lesson 3 content)? If so, explicitly reference one of those question cards at the start of Lesson 3 to reinforce the coherent storyline. (5) Was the 40-minute lesson long enough to complete all five phases, including the die-rolling experiment? If time was short, consider having the token-bag observation as a homework extension or a 5-minute starter in Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners rolled a six-faced die 60 times per group and drew tokens from a bag of githeri-coloured tokens (white/maize, brown/beans, green/peas), recording all outcomes in tally tables and computing experimental probabilities as fractions, decimals, and percentages.
What did I learn?	Learners derived the theoretical probability formula $P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes}$, and applied it to Kenyan contexts: selecting a name from a class of 30 learners, and drawing a coloured token from a mixed bag. They confirmed that all individual outcome probabilities for a fair die sum to 1.
How does this explain the phenomenon?	Learners explained, using the probability formula and the concept of equally likely outcomes, why Gor Mahia FC's past winning streak does not change the probability of the next match result — directly challenging the gambler's fallacy promoted in Nairobi betting advertisements.

LESSON 3 (80 minutes (2 × 40-minute lessons)): Range of Probability: From Impossible to Certain

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate and formalise the boundaries of probability (0 to 1), placing real-life Kenyan events on a probability number line and expressing probability using fractions, decimals, and percentages — building the mathematical foundation needed to dismantle the gambler's fallacy from the anchoring phenomenon.
Knowledge	Learners understand that the probability of any event must satisfy $0 \leq P(\text{event}) \leq 1$, where 0 represents an impossible event and 1 represents a certain event; they can express probability as a fraction, decimal, and percentage.
Skills	Learners can: (a) determine the range of probability of an event; (b) place events on a probability number line; (c) convert probability expressions between fractions, decimals, and percentages; (d) justify why $P > 1$ or $P < 0$ is mathematically impossible.
Attitudes	Learners appreciate that probability gives precise, bounded mathematical values to uncertainty, and begin to question culturally promoted misconceptions — such as gambling advertisements on Nairobi matatus claiming one can 'beat the odds' — using this mathematical reasoning.
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	In Lessons 1 and 2 learners established what equally likely events are and computed basic probabilities for coin tosses and dice rolls. In Lesson 3 they now ask: what are the mathematical limits of any probability? By anchoring this in vivid Kenyan scenarios — the certainty of sunrise over Mombasa versus the impossibility of snow in Kisumu in July — learners construct a probability number line (0 to 1) and formalise the range rule. This directly equips them to challenge the gambler's fallacy: if each match is independent and all probabilities are bounded between 0 and 1, no 'streak' can push a future probability outside those bounds. The DQB is updated to reflect this new constraint.
Safety Notes	No physical hazards. Sensitivity note: when discussing gambling advertisements, affirm that no learner or family member is being judged; frame the discussion as mathematical analysis of advertising claims, not personal criticism.

B. LESSON OVERVIEW

Lesson 3 formalises the range of probability ($0 \leq P \leq 1$) through a structured Predict–Observe–Explain–DQB–Model sequence. Learners first predict where they would place eight Kenyan real-life events on an unmarked line from 'impossible' to 'certain'. They then observe and analyse evidence — a short Kenyan scenario card set and a class coin-toss frequency table accumulated in Lesson 1 — to test their predictions. In the Explain phase, learners formalise that no experiment can yield $P > 1$ or $P < 0$, convert between fraction, decimal, and percentage representations, and connect this directly back to Gor Mahia's winning streak: no matter how long the streak, the probability of winning or losing the next match remains bounded between 0 and 1, never becoming 'inevitable'. Learners update the Driving Question Board and revise their cumulative probability model to include the 0-to-1 constraint boundary.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Each learner receives a strip of	Distribute Event Cards and blank	Elicitation of prior knowledge via	Circulate during individual work.

 VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: probability range impossible event  EXERCISE: Making predictions with probability Related topic: not an exact match for this lesson. Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	paper representing a blank number line labelled 'Impossible' at the left end and 'Certain' at the right end. They are given a set of eight Event Cards (printed or read aloud) describing Kenyan scenarios, e.g.: (1) The sun rises tomorrow over Mombasa. (2) It snows in Kisumu in July. (3) A flipped 10-shilling coin lands on heads. (4) Gor Mahia wins their very next Premier League match. (5) A matatu from Nairobi to Nakuru takes exactly 2 hours with no traffic. (6) A student chosen at random from a class of 40 is a girl, if 20 are girls. (7) A ripe mango falls upward when dropped. (8) It rains at least once in Nairobi in April. Learners individually place each event on their number line using a letter code (A–H), then write a one-sentence justification for each placement. They do NOT yet assign numerical values.	number line strips. Say: 'Before we do any mathematics today, I want to see what your instincts tell you. Place each event somewhere on this line — do not worry about exact numbers yet. You have 5 minutes, working alone.' Allow 5 minutes of silent individual work. After 5 minutes, say: 'Turn to your neighbour and compare your placements. Do you agree? If not, try to convince each other using one reason.' Allow 3 minutes of pair discussion. Cold-call 4 learners (choose one confident, one hesitant, one who seems to have placed an event differently from peers, and one who has written a detailed justification): 'Tell me where you placed Event 2 — snow in Kisumu — and why.' After each response, wait 12 seconds before probing: 'Does anyone place this differently? Why?' Do NOT confirm or correct yet. Record the class's most common placements on the board using tally marks next to each event label to create visible disagreement.	anchored prediction: by forcing placements on a physical continuum before instruction, learners activate intuitive understanding of likelihood. Recording disagreements publicly creates cognitive dissonance that motivates the Observe phase. Connecting events to vivid Kenyan contexts (Kisumu snow, Nairobi April rain, Gor Mahia match) makes the abstraction of 'range' personally meaningful.	Look for: (a) learners who place 'impossible' and 'certain' events in the middle of the line — flag for targeted questioning in Phase 3; (b) learners who correctly distinguish near-certain (Nairobi rain in April) from certain (sunrise) — note as potential peer explainers. Collect 4–5 strips at random to review justifications before the Explain phase.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: probability range impossible event  EXERCISE: Making predictions with probability Related topic: not	Activity A — Scenario Card Analysis (15 min): In groups of 4, learners receive a set of 6 'Evidence Cards'. Each card presents a Kenyan scenario with a completed sample space, e.g.: Card 1 — 'A bag contains 3 red and 3 blue maize kernels. One is picked at random.' Sample space = {R, R, R, B, B, B}. Favourable outcomes for 'picking red' = 3. Total outcomes = 6. Learners are asked to write the probability as a fraction, then convert it to a decimal and a percentage. Card 2 — 'A student randomly picks one fruit from a bowl containing only	Distribute Evidence Cards to each group. Say: 'Your job is to be mathematical detectives. For each card, calculate the probability, then write it three ways: as a fraction, as a decimal, and as a percentage. Then place it on your number line. You have 12 minutes.' Set a visible timer. Circulate — do NOT give answers. Use prompt questions: 'How many outcomes are in the sample space?' (wait 10 seconds), 'How many of those are favourable?' (wait 10 seconds), 'What does it mean if the number of favourable outcomes equals zero?' (wait 15	Concrete-to-abstract bridging using graduated Evidence Cards: Cards begin with physically tangible Kenyan objects (maize kernels, fruit bowls) before moving to abstract spinners, so all learners access the mathematics. The impossible and certain cards (only mangoes, picking guava) create the 0 and 1 anchor points experientially before they are formalised. Connecting to the class's own accumulated coin-toss data from Lesson 1 reinforces continuity of evidence across the storyline.	Walk the room during group work. Observable indicators of understanding: (a) learners correctly write $P = 0$ for the 'impossible' card and $P = 1$ for the 'certain' card; (b) learners successfully convert fractions to decimals and percentages without prompting; (c) learners place all six events between or at 0 and 1 on the number line. Observable indicators of confusion: (d) learners write $P = 6/6 = 1$ but do not connect this to 'certain'; (e) learners cannot convert fractions to decimals — note for targeted support in Explain phase. Check at

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	mangoes. What is the probability of picking a mango?' Card 3 — 'A student randomly picks one fruit from a bowl containing only mangoes. What is the probability of picking a guava?' Cards 4–6 escalate: a six-faced die, a football match with 3 outcomes (win/draw/lose), and a spinner divided into 5 equal sections. For each card, learners record: fraction, decimal, percentage, and position on their number line. Activity B — Accumulated Coin-Toss Data (5 min): The teacher displays the class's running coin-toss frequency table built in Lesson 1 (total tosses ≈ 200+ across the class). Learners calculate the experimental probability of heads as a fraction, decimal, and percentage, then compare to the theoretical value of 0.5.	seconds — this targets Cards 2 and 3 for impossible/certain recognition). After 12 minutes, call the class back. Display the accumulated coin-toss data on the board. Ask: 'Look at our data from the last two lessons. What is the experimental probability of heads as a decimal? As a percentage?' Cold-call 2 learners. Then ask: 'Is this value closer to 0 or to 1, or somewhere in between?' (wait 12 seconds). Ask: 'Did any group get a probability fraction where the numerator was larger than the denominator?' (pause 10 seconds). 'What would that even mean?'	least 3 groups' written work before transitioning.	
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: probability range impossible event  EXERCISE: Making predictions with probability <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Part A — Formalising the Number Line (10 min): Each learner now draws a formal probability number line in their exercise books, labelling 0 (Impossible) and 1 (Certain), and marking 0.5 ($\frac{1}{2}$, 50%) at the midpoint. They revisit their Event Cards from Phase 1 and now assign numerical values (fractions, decimals, percentages) to each event, correcting their original placements where needed. They write a formal probability statement for at least 4 of the 8 events, e.g., 'P(sun rises over Mombasa tomorrow) = 1' and 'P(snow in Kisumu in July) = 0'. Part B — The Impossibility of $P > 1$ and $P < 0$ (8 min): Learners read a short 2-paragraph printed or on-screen text: a fictional betting advertisement claiming 'Our algorithm gives you a 120%	Open Part A by saying: 'Now we are going to make your intuitions precise. Open your books to a fresh page. Draw a line — label the left end 0 and write Impossible, label the right end 1 and write Certain. Mark 0.5 in the middle.' Give learners 2 minutes to draw and label. Then say: 'Go back to your Event Cards. This time, I want a number — not just a position. Write $P(\text{event}) = [\text{fraction}] = [\text{decimal}] = [\text{percentage}]$ for at least four events.' Allow 5 minutes. Cold-call 3 learners to share one event each — choose deliberately: one certain, one impossible, one middle-ground. After each, ask the class: 'Agree or disagree? Thumbs up or down. Why?' Wait 10 seconds after each thumb response before elaborating. For Part B, distribute or display the	Three-part explain sequence using claim-evidence-reasoning (CER): learners first build the formal number line (claim), then use the betting advertisement to construct a refutation (evidence + reasoning), then apply reasoning directly to the phenomenon (transfer). The CER structure is an NGSS Science and Engineering Practice (Constructing Explanations) applied to mathematics. Returning to the phenomenon in Part C ensures every phase of the lesson ties back to the storyline and driving question.	Collect exercise books from 6 learners (2 high, 2 middle, 2 struggling based on prior lessons) at the end of Part A to check: (a) correct numerical values assigned to certain and impossible events; (b) correct fraction-decimal-percentage conversions. During Part B, listen for the key phrase 'probability cannot exceed 1' or equivalent — if fewer than half the class produces this in their written responses, pause and reteach using the maize kernel example before proceeding to Part C. In Part C, the written response about Gor Mahia serves as an exit check for the core SLO: learners must reference $0 \leq P \leq 1$ explicitly.

	chance of predicting the winner!' Learners are asked: 'What is mathematically wrong with this claim? Use what you know about the range of probability to explain in two sentences.' They write their response individually, then share with a partner. Part C — Connecting to the Phenomenon (5 min): The teacher returns to the Gor Mahia phenomenon. Learners answer in their books: 'A sports commentator says after Gor Mahia's 8th consecutive win: the probability of them winning the next match has now risen above 1 because they are on such a hot streak. Using the range of probability, explain why this commentator is mathematically wrong.'	fictional advertisement. Say: 'A company is advertising gambling odds. Read this carefully. What is wrong — mathematically wrong — with their claim? You have 3 minutes to write your answer alone.' After 3 minutes: 'Share with your partner. Decide on the best one-sentence mathematical argument between you.' Cold-call 2 pairs. After responses, say firmly: 'Probability can never exceed 1. It can never be less than 0. This is not an opinion — it is a mathematical law. If anyone ever shows you odds above 100%, they are either making a mistake or deliberately misleading you.' For Part C, say: 'Now apply this directly to Gor Mahia. Write your answer in your book. Two sentences minimum.' Allow 4 minutes. Cold-call 2 learners. Conclude: 'No winning streak — not Gor Mahia's, not anyone's — can push the probability of the next event above 1 or below 0. The mathematics does not allow it.'		
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: probability range impossible event  EXERCISE: Making predictions with probability Related topic: not	Learners return to the class Driving Question Board. They individually write one sticky note (or a slip of paper) responding to the prompt: 'What does today's lesson add to our answer to the driving question: If past outcomes do not change future probabilities, how can mathematics protect us from gambling?' They must reference the range of probability (0 to 1) in their response. Learners also scan the existing DQB cards from Lessons 1 and 2 and identify one question that today's lesson has now partially or fully answered. They mark it with a tick and write a brief explanation on the reverse side. A class scribe	Say: 'Look at our Driving Question Board. We have been building an answer across three lessons. Today's piece is about limits — mathematical boundaries. On your sticky note, explain how knowing that probability must stay between 0 and 1 helps us respond to the driving question. You have 3 minutes.' After writing, say: 'Now scan the board. Find one question from Lesson 1 or 2 that you can now answer — or partially answer — because of what you learned today. Mark it and write your answer on the back. 2 minutes.' Call the class scribe to read out 3–4 new notes. After each, ask: 'Does this advance our driving	Progressive DQB construction as a visible thinking routine: by physically marking previously posted questions as 'answered', learners see their own growing understanding made tangible. The voting mechanism creates collaborative epistemic accountability — the class, not just the teacher, evaluates the quality of reasoning. Connecting the DQB update to the phenomenon and previewing Lesson 4 maintains storyline coherence.	Read all sticky notes after class. Look for: (a) learners who correctly cite '$0 \leq P \leq 1$' as the constraint — these learners have met the SLO; (b) learners who write only vague statements like 'probability is between 0 and 1' without connecting to gambling or the phenomenon — provide written feedback prompting them to make the connection explicit; (c) learners who cannot identify any previously answered question — flag for individual check-in at the start of Lesson 4.

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(rotating role) reads out 3–4 new sticky notes and the class votes (show of hands) on which one best advances the driving question.	question? Thumbs up or sideways.' After voting, say: 'Pin the strongest ones near the top of the board — they are moving us closer to the answer.' End by framing the next lesson: 'We now know probabilities are bounded. In Lesson 4, we ask: what happens when two events cannot both occur at the same time — like Gor Mahia both winning and losing the same match?'		
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: probability range impossible event  EXERCISE: Making predictions with probability <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners open their cumulative probability model (begun in Lesson 1 as a hand-drawn or digital diagram). In Lesson 1 they added equally likely events and sample spaces; in Lesson 2 they added the probability formula $P = \text{favourable outcomes} / \text{total outcomes}$. Today they add a new layer: a clearly drawn and labelled probability number line (0 to 1) with at least four Kenyan events placed on it. They annotate the model with: (a) a boundary rule box stating '$0 \leq P(\text{any event}) \leq 1$ — this is always true'; (b) an arrow pointing from the Gor Mahia winning streak scenario to the number line with the annotation 'no streak changes the boundaries'; (c) a colour-coded legend distinguishing impossible (red, $P=0$), certain (green, $P=1$), and intermediate (blue, $0 < P < 1$) events. Learners share their updated model with a shoulder partner for 2 minutes of peer feedback, using the prompt: 'Is the number line correctly labelled? Are the Kenyan events placed accurately? Is the boundary rule written clearly?'	Say: 'Take out your cumulative probability model. We are adding today's learning to it. Your model should now show three things we have learned across three lessons: sample spaces, the probability formula, and now the range. You have 8 minutes to update it.' Circulate — do NOT draw the model for any learner, but use targeted questions: 'Where does $P = 0$ sit on your number line?' (wait 10 seconds); 'Can you find a Kenyan event to place at exactly 0.5?' (wait 12 seconds); 'What does your annotation say about Gor Mahia's streak?' (wait 10 seconds). After 8 minutes: 'Share with your shoulder partner. Use the three-question peer feedback prompt on the board. You have 2 minutes.' After peer feedback: 'Make any corrections. Your model is a living document — it will grow every lesson.' Conclude the lesson by saying: 'Three lessons in, we have built a foundation: probability is a number bounded between 0 and 1. No superstition, no streak, no betting algorithm can change that mathematical fact. Hold onto that as we continue.'	Cumulative model revision as an NGSS Developing and Using Models practice: each lesson's model update forces learners to integrate new knowledge with prior knowledge rather than treating each lesson as isolated. The colour-coding system creates a visual schema that persists across all 8 lessons. Peer feedback using a structured 3-question prompt builds metacognitive awareness and communication competency as mandated by KICD core competencies.	Conduct a gallery walk of 6–8 models (quickly) while learners do peer feedback. Look for: (a) probability number line correctly anchored at 0 and 1 with accurate event placements — this is the core SLO indicator; (b) the boundary rule written in the learner's own words (not copied verbatim) — indicates genuine understanding; (c) the Gor Mahia annotation connecting the phenomenon to the range rule — indicates storyline transfer. Any model missing the boundary rule annotation should receive a written question: 'What happens to P if the numerator of your fraction is larger than the denominator? Can that ever happen?' to prompt self-correction before Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully distinguish between 'near-certain' events (e.g., rain in Nairobi in April) and 'certain' events (e.g., sunrise over Mombasa) — if not, what Kenyan examples would make this distinction sharper in Lesson 4's review? (2) During Part B of the Explain phase, how many learners independently used the language ' $0 \leq P \leq 1$ ' without being prompted — if fewer than 60%, consider opening Lesson 4 with a 5-minute retrieval warm-up using two more betting-advertisement claims to reactivate the boundary rule. (3) Were there learners who struggled with the fraction-to-decimal-to-percentage conversion? If so, note their names and plan a brief bridging task at the start of Lesson 4 using familiar Kenyan currency (e.g., 50 cents out of 1 shilling = $0.5 = 50\%$) to scaffold the conversion before the lesson proceeds to mutually exclusive events. (4) Did the DQB sticky notes show genuine engagement with the gambling/phenomenon connection, or were they procedural restatements of the rule? If the latter, consider adding a 'so what?' sentence-starter prompt ('This matters for our driving question because...') to the DQB protocol in Lesson 4. (5) How is the cumulative model developing across the class — are learners treating it as a meaningful record of growing understanding, or as a task to complete? If the latter, consider projecting one strong model anonymously at the start of Lesson 4 and discussing what makes it a good thinking tool.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners analysed Kenyan scenario Evidence Cards (maize kernels, fruit bowls, dice, spinners) and the class's accumulated coin-toss data from Lesson 1 to find probabilities and place them on a number line.
What did I learn?	The probability of any event must satisfy $0 \leq P(\text{event}) \leq 1$; $P = 0$ means impossible (e.g., snow in Kisumu in July); $P = 1$ means certain (e.g., sunrise over Mombasa); all other events fall strictly between 0 and 1, and no winning streak — including Gor Mahia's — can push a future probability outside these bounds.
How does this explain the phenomenon?	Learners used the range of probability to refute a fictional gambling advertisement claiming '120% odds', and wrote a mathematical explanation of why a sports commentator claiming Gor Mahia's winning streak raises their probability above 1 is mathematically impossible, using the boundary rule $0 \leq P \leq 1$.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Mutually Exclusive Events — Identification & The Addition Rule

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners will identify mutually exclusive events in real-life Kenyan contexts and calculate the probability of mutually exclusive events using the addition rule $P(A \text{ or } B) = P(A) + P(B)$, connecting this understanding to the anchoring phenomenon of the Gambler's Fallacy and the dangers of gambling.
Knowledge	Learners will know that: (a) mutually exclusive events are events that cannot occur at the same time; (b) for mutually exclusive events, $P(A \text{ or } B) = P(A) + P(B)$; (c) the addition rule applies only when events share no outcomes ($P(A \text{ and } B) = 0$); (d) many gambling predictions falsely treat non-mutually-exclusive outcomes as certain, exploiting misunderstanding of the addition rule.
Skills	Learners will be able to: (a) identify whether two events are mutually exclusive or not from a description or experiment; (b) sort a set of events into 'mutually exclusive' and 'not mutually exclusive' categories with justification; (c) calculate $P(A \text{ or } B)$ using the addition rule for mutually exclusive events from single-chance experiments (one die, one card draw, one raffle draw); (d) explain why the addition rule cannot be blindly applied to non-mutually-exclusive events.
Attitudes	Learners will appreciate that: (a) precise mathematical language prevents costly errors — including in gambling decisions; (b) working collaboratively to sort and justify event classifications builds shared mathematical understanding; (c) probability mathematics protects communities from exploitation by gambling advertisers who blur distinctions between mutually exclusive outcomes.
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	In Lessons 1–3, learners established equally likely events, the probability range $[0,1]$, and the independence of each match or coin toss from past results. Lesson 4 now asks: when we combine outcomes — 'What is the probability that Gor Mahia wins OR draws this match?' — how do we add probabilities correctly? Learners discover that the addition rule only works cleanly when the two outcomes cannot both happen at once (mutually exclusive). This directly addresses gambling ads on Nairobi matatus that imply bettors can simply 'add up' their chances across multiple simultaneous selections — a mathematically flawed shortcut that costs bettors money.
Safety Notes	No physical hazards. Sensitivity note: some learners may have family members affected by gambling addiction. The teacher should frame all gambling-related discussions around mathematical literacy and protection, never judgment of individuals. Use 'gambling advertisements exploit misunderstanding of probability' framing, not 'gamblers are foolish.'

B. LESSON OVERVIEW

Learners open by revisiting the Gor Mahia / betting-advertisement phenomenon and are challenged with a new question: 'If a bettor picks Gor Mahia to win AND to draw the same match, can those both happen? What is the probability of one OR the other?' Through a card-sorting activity using Kenyan context event cards, learners classify events as mutually exclusive or not. They then conduct a structured die-rolling and raffle-simulation experiment to verify the addition rule $P(A \text{ or } B) = P(A) + P(B)$ for mutually exclusive events. Guided examples anchor the rule to the school raffle and football match scenarios. Learners update their Driving Question Board and revise their cumulative probability model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Mutually Exclusive Events Principles Source: CK-12  Search ARES for similar videos  HTML: Independent, Conditional, and Mutually Exclusive Events (Flexbook) Source: CK-12  Search ARES for similar readings	Learners are shown a slide/board prompt: 'A sports betting slip shows two selections for a SINGLE Gor Mahia match: (A) Gor Mahia WINS, (B) Gor Mahia DRAWS. The advertiser says: your chance of being right is $P(\text{Win}) + P(\text{Draw}) = 1/3 + 1/3 = 2/3$. Is this claim correct? Can both A and B happen on the same match? Write your prediction and reason in your exercise book — do NOT discuss yet.' Learners write individually for 3 minutes, then share with a shoulder partner for 2 minutes.	Display the prompt on the board or ARES screen. Say: 'Before we investigate anything today, I want your honest prediction. There is no wrong answer yet — I want to see your thinking.' Give WAIT TIME of 3 minutes of silent writing. Circulate and note 3–4 different responses (one who says 'yes, add them,' one who says 'no, they can't both happen,' one who is unsure). After partner share, cold-call exactly 3 learners: 'Amina, what did you predict and why?' / 'Ochieng, do you agree or disagree with Amina?' / 'Wanjiru, what did your partner say that surprised you?' Record key prediction phrases verbatim on the board under the heading 'OUR PREDICTIONS — we will test these today.'	Elicit-and-Record: capturing raw predictions publicly creates cognitive dissonance when evidence later contradicts them, motivating genuine inquiry. Connecting to the betting-advertisement phenomenon ensures learners see immediate real-world relevance.	Observable indicator: every learner has a written prediction with a reason in their book before discussion begins. Teacher scans books during circulation — look for whether learners use the word 'same match' or 'at the same time' as a natural constraint; this reveals prior intuition about mutual exclusivity. Note which learners conflate 'adding probabilities' with 'combining events' — these learners need targeted questioning in Phase 3.
Observe Phase  VIDEO: Mutually Exclusive Events Principles Source: CK-12  Search ARES for similar videos  HTML: Mutually Exclusive Events (Flexbook) Source: CK-12  Search ARES for similar readings	ACTIVITY A — Event Card Sort (15 min): Each group of 4 receives a set of 10 event-cards (printed or written on cut paper). Each card describes two events for the SAME single trial. Groups physically sort cards into two piles: 'CAN happen at the same time' and 'CANNOT happen at the same time.' Event cards include: (1) Rolling a die: getting a 3 AND getting a 5; (2) Rolling a die: getting an even number AND getting a number greater than 4; (3) Drawing one card: drawing a Heart AND drawing a Spade; (4) Drawing one card: drawing a King AND drawing a Heart; (5) A matatu from Nairobi CBD: arriving in Mombasa AND arriving in Kisumu at the same time; (6) A learner's Grade 10 exam result: scoring an A AND scoring a D on the same	Distribute event cards to groups. Say: 'Your job is simple — for each card, ask: on ONE single trial, can BOTH things happen at exactly the same time? If yes, one pile. If no, other pile. Do not use any vocabulary I have taught you yet — use your own words.' Give groups 10 minutes. Circulate and listen — do NOT correct yet. Note groups who correctly identify card (4) King AND Heart as 'CAN happen' — this is the key diagnostic. After sorting, say: 'Keep your piles. We will come back to them.' Transition to die-rolling: 'Now let us collect actual data.' Distribute dice. Say: 'Roll 30 times. Each time, record what you got. We want to know: how often do you get a 2? How often a 5? How often either a 2 or a 5?' After 10 minutes, call table	Concrete Manipulation + Data Collection: the physical card sort activates embodied reasoning about simultaneity before formal vocabulary is introduced. The die experiment provides empirical evidence that $P(A \text{ or } B) \approx P(A) + P(B)$ for these events, making the later algebraic rule feel discovered rather than imposed. Pooling class data reduces sampling error and models scientific practice.	Observable indicator 1: each group's card-sort correctly places cards (1), (3), (5), (6), (7), (8), (10) in 'CANNOT happen' and cards (2), (4), (9) in 'CAN happen.' Card (4) — King AND Heart — is the key diagnostic: groups who place it in 'CANNOT happen' reveal a misconception that different categories are always mutually exclusive; flag these groups for targeted questioning in Phase 3. Observable indicator 2: in the die experiment, learners record $P(2 \text{ or } 5)$ as a value close to but not equal to $P(2) + P(5)$ due to sampling variation — check that learners interpret this as 'close to' not 'exactly,' building realistic data-literacy expectations.

	paper; (7) Gor Mahia's match result: winning AND drawing the same match; (8) Picking one ball from a bag of red, blue and green balls: picking red AND picking blue; (9) Picking one ball: picking red AND picking a ball that is NOT green; (10) A raffle ticket: winning 1st prize AND winning 2nd prize in a raffle where one ticket wins one prize only. Groups record their sorted lists and write a one-sentence justification for each. ACTIVITY B — Die-Roll Experiment (10 min): Each pair rolls a standard die 30 times. They record: frequency of rolling a 2 (Event A), frequency of rolling a 5 (Event B), frequency of rolling a 2 OR a 5 (Event A or B). They compute experimental probabilities. Class results are pooled on the board into a tally table.	representatives to report frequencies. Record class totals on the board. Calculate class-level experimental $P(2) \approx 1/6$, $P(5) \approx 1/6$, $P(2 \text{ or } 5) \approx 2/6$. Give WAIT TIME of 15 seconds after writing the three values and ask: 'What pattern do you notice about these three numbers?' Cold-call 2 learners before explaining.		
Explain Phase  VIDEO: Die rolling probability with independent events - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Mutually Exclusive Events Practice Source: CK-12  Search ARES for similar readings	STEP 1 — Vocabulary Anchoring (5 min): Teacher introduces the formal term 'mutually exclusive events' and links it directly to learners' own card-sort language. Learners re-label their two piles using the new term. STEP 2 — Addition Rule Derivation (10 min): Teacher works through three guided examples on the board, using the school-raffle and football-match contexts. Example 1 (raffle): 'A school raffle at Nakuru High School has 50 tickets. Mwangi holds ticket 7. Event A = Mwangi wins 1st prize (one winner). Event B = Mwangi wins 2nd prize (one winner, different ticket). Are A and B mutually exclusive? $P(A) = 1/50$. $P(B) = 1/50$. $P(A \text{ or } B) = ?$' Learners calculate individually, then teacher confirms $P(A \text{ or } B) = 2/50 = 1/25$.	For Step 1, point to the 'CANNOT happen at the same time' pile and say: 'Mathematicians have a precise name for exactly what you described — events that CANNOT occur simultaneously on a single trial are called MUTUALLY EXCLUSIVE events. Write this definition in your own words beneath the one I will write.' Write on board: 'Two events are mutually exclusive if they cannot both occur in a single trial. $P(A \text{ and } B) = 0$.' Give WAIT TIME of 15 seconds. For Step 2, after writing Example 1, say: 'Do not watch me — calculate $P(A \text{ or } B)$ yourself first. 30 seconds.' Cold-call 1 learner for the answer. Then ask: 'How did the addition rule appear in your die experiment from Activity B? Did it hold exactly?' Cold-call 2 learners. For Step 3 (gambling	Bridging from Concrete to Formal: the explicit connection from learners' own card-sort language to the formal definition creates ownership of the vocabulary. The contrast case (King AND Heart) is essential — without a non-mutually-exclusive example, learners over-generalise the addition rule. The gambling advertisement deconstruction uses the 'Claim-Evidence-Reasoning' framework: learners identify the claim, test it with their probability tools, and reason about the advertiser's intent.	Observable indicator 1: during Example 3, all learners should be able to name the King of Hearts as the counter-example that makes WIN a King AND a Heart non-mutually exclusive — scan the room for written notes confirming this. Observable indicator 2: during the gambling deconstruction, listen for learners who can spontaneously connect back to Lesson 3 (independence) — these learners are demonstrating cross-lesson integration. Red flag: any learner who says '$P(\text{Win or Draw}) = 2/3$ means you'll win 2 out of every 3 bets' is conflating probability with certainty — address immediately with: 'Does probability tell us what WILL happen, or what is LIKELY over many trials?'

	Example 2 (die): 'Roll one die. P(rolling a 2 or a 5) = ?' Links to their experimental data. Example 3 (non-mutually exclusive, contrast): 'Draw one card from a standard deck. Event A = King. Event B = Heart. Are these mutually exclusive?' — Teacher deliberately returns to card (4) from the sort. Learners debate: 'Can a card be BOTH a King AND a Heart? Yes — King of Hearts exists!' Therefore $P(A \text{ or } B) \neq P(A) + P(B)$. Teacher states: 'Today we only apply the addition rule when events ARE mutually exclusive.' STEP 3 — Gambling Advertisement Deconstruction (8 min): Teacher displays a mock betting-slip advertisement (Nairobi matatu style): 'Pick Gor Mahia to WIN or DRAW — 2 chances out of 3! Your odds are great!' Learners work in pairs: 'Is WIN and DRAW mutually exclusive for ONE match? So can we use $P(\text{Win or Draw}) = P(\text{Win}) + P(\text{Draw})$? What does this actually equal if each outcome has probability $1/3$?' Learners calculate $P(\text{Win or Draw}) = 2/3$. Teacher asks: 'So is the advertiser's maths correct here? Why might the advertiser WANT you to focus on this number instead of $P(\text{Lose}) = 1/3$?' Learners discuss the psychological and mathematical manipulation.	deconstruction), say: 'Work with your partner. I want two things: first, confirm whether WIN and DRAW are mutually exclusive for one match. Second, tell me whether the advertiser's claim is mathematically correct AND whether $2/3$ is the full picture.' After 3 minutes, cold-call 3 pairs. Conclude by saying: 'The advertiser is technically correct about the addition — but they are hiding $P(\text{Lose}) = 1/3$ and they are ignoring that across MANY bets, independent losses accumulate. This is what Lessons 1–3 showed us about independence.'		
Driving Question Board (DQB) Creation  VIDEO: Die rolling probability with independent events <i>Related topic: not</i>	Each learner reviews the class Driving Question Board (established in Lesson 1): 'If past outcomes of a chance event do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling?' Learners individually	Distribute sticky notes or direct learners to the DQB journal page. Say: 'You have 4 minutes. Two things only: one thing you now understand better about our driving question, and one new question today's lesson raised for you.' After 4 minutes, cold-call 3 learners to share their new questions. Write the new questions	Metacognitive Monitoring via DQB: the Driving Question Board functions as a public, cumulative record of the class's growing understanding. By requiring both an 'understanding gained' and a 'new question raised,' the protocol reinforces that science and mathematics are iterative — new knowledge always generates new	Observable indicator: learners' 'understanding' statements should reference at minimum ONE of the following: (a) the definition of mutually exclusive events, (b) the addition rule, or (c) a Kenyan-context example from today. Learners whose statements are vague ('I learned about probability') need a follow-up

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Independent, Conditional, and Mutually Exclusive Events (Flexbook) Source: CK-12 Search ARES for similar readings	write on a sticky note or in their ARES journal: (a) ONE new understanding from today that advances the driving question — e.g., 'I can now calculate the probability of Gor Mahia winning OR drawing using the addition rule, but only because those outcomes can't happen simultaneously'; (b) ONE new question the day's learning raised — e.g., 'What happens to the addition rule when events CAN overlap, like drawing a King or a Heart?' The class adds new question-notes to the DQB under the heading 'Mutually Exclusive Events.' The teacher reads 3–4 notes aloud and connects them to the lesson sequence: 'Notice how Lesson 3 told us independence means past results don't matter. Today's lesson tells us that when outcomes are mutually exclusive, we can ADD their probabilities for a single trial. These are two different tools.'	on the DQB physically or on the ARES platform board. Point to the existing DQB entries from Lessons 1–3 and say: 'Look at the question from Lesson 2 about whether we can predict a coin landing on heads. How does today's lesson — about adding mutually exclusive probabilities — connect to that earlier question?' Give WAIT TIME of 15 seconds. Cold-call 2 learners. Affirm and clarify: 'Each lesson is giving us one more mathematical tool. The driving question asks us to use ALL of them together.'	questions. Connecting current DQB entries to prior-lesson entries builds the narrative arc of the 8-lesson sequence.	prompt: 'Can you name a specific example from today that showed you this?' New questions appearing on the DQB that anticipate future lessons (e.g., about overlapping events, or about multiple bets) indicate learners who are ready for extension.
Model Building Phase  VIDEO: Addition rule for probability <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Mutually Exclusive Events (Flexbook) Source: CK-12 Search ARES	Learners retrieve their cumulative probability model from Lesson 3 (a diagram or annotated framework showing: equally likely events, probability range [0,1], independence of events across trials). They now ADD a new layer to the model specifically for a single trial: a branching diagram showing one die roll with all six outcomes. They annotate: (a) circle or bracket events that ARE mutually exclusive (e.g., {rolling a 2} and {rolling a 5}); (b) write $P(2 \text{ or } 5) = P(2) + P(5) = 1/6 + 1/6 = 2/6 = 1/3$ alongside; (c) annotate the Gor Mahia match branch: outcomes WIN / DRAW / LOSE — all three are mutually exclusive for one match, so $P(\text{Win or Draw}) =$	Say: 'Take out your probability model from Lesson 3. Today you are adding a new layer — the addition rule for mutually exclusive events on a SINGLE trial. This is separate from what we learned about independence, which was about MULTIPLE trials over time.' Give learners 8 minutes to annotate. Circulate and check that: (1) the branching die diagram clearly shows all six outcomes labelled; (2) mutually exclusive event pairs are visually bracketed or circled; (3) the addition rule is written as a formula, not just a number; (4) the RED WARNING BOX about non-mutually-exclusive events is present. After 8 minutes, say: 'Turn to your partner. Point to	Cumulative Model Revision: requiring learners to physically annotate an ongoing model — rather than redraw it each lesson — makes conceptual growth visible and tangible. The RED WARNING BOX is a deliberate metacognitive device: it forces learners to encode the boundary condition of the rule, preventing over-generalisation. The partner explanation (verbal model) reinforces encoding through the dual-coding of visual and linguistic representation.	Observable indicator 1: the completed model diagram includes all four required components (branching diagram, mutually exclusive brackets, addition rule formula, red warning box). Use this as a 2-minute observation checklist while circulating. Observable indicator 2: during partner explanation, listen for correct use of the phrase 'single trial' — learners who omit this qualifier may be conflating the addition rule with multi-trial calculations. Observable indicator 3: the red warning box demonstrates awareness of the rule's boundary condition — learners who include a specific example (King of Hearts) in the

for similar readings	$1/3 + 1/3 = 2/3$; (d) add a RED WARNING BOX: 'Addition rule for mutually exclusive events ONLY — does NOT apply when outcomes can overlap (e.g., King AND Heart).' Learners share their updated model with a partner and explain one annotation aloud before the lesson ends.	ONE part of your model and explain it out loud using the words: mutually exclusive, addition rule, and single trial. You have 2 minutes.' Cold-call 2 pairs to share. Close by connecting to the phenomenon: 'When the betting advertisement on the matatu says Gor Mahia will WIN or DRAW — they are using the addition rule correctly for one match. But as we proved in Lesson 3, knowing the probability of one match tells you nothing about whether they will win the NEXT one. That is the gambler's fallacy — and it is how betting companies make their money.'	box show deeper understanding than those who write only a general statement.
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) CARD SORT DIAGNOSTIC — Did most groups correctly identify card (4) — King AND Heart — as NOT mutually exclusive? If groups placed it in the 'CANNOT happen' pile, the next lesson must revisit overlapping events before proceeding to independent events. Note the proportion of groups who got this right without prompting. (2) ADDITION RULE OWNERSHIP — During the guided examples, did learners calculate $P(A \text{ or } B)$ before you showed them, or did they wait to copy? If learners waited, consider restructuring Phase 3 next time to require written individual working before any class discussion. (3) GAMBLING DECONSTRUCTION — Did the Gor Mahia advertisement discussion remain focused on mathematical critique, or did it drift into general opinions about gambling? If it drifted, prepare a tighter prompt for Lesson 5: 'We are mathematicians analysing claims, not judges of people's choices.' (4) DQB QUALITY — Review the sticky notes added to the DQB. Are learners' new questions getting more mathematically sophisticated across lessons (from 'what is probability?' in Lesson 1 to 'what about overlapping events?' in Lesson 4)? If questions are not advancing, the learners may not be reading prior DQB entries — build in a 2-minute silent DQB review at the start of Lesson 5. (5) MODEL COHERENCE — Do learners' updated models clearly separate the concept of independence (across multiple trials, first met informally in Lesson 1) from mutual exclusivity (within a single trial, introduced today)? Confusion between these two concepts is the most common error at this point in the sub-strand and must be resolved before Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners sorted 10 Kenyan-context event-cards into 'mutually exclusive' and 'not mutually exclusive' piles, justifying each placement. They also rolled a die 30 times each and recorded experimental frequencies for Event A (rolling a 2), Event B (rolling a 5), and Event A or B (rolling a 2 or a 5), pooling class data on the board.
What did I learn?	Mutually exclusive events are events that cannot occur simultaneously on a single trial — for example, a matatu cannot arrive in Nairobi and Mombasa at the same time; a Gor Mahia match cannot end in both a WIN and a DRAW. For mutually exclusive events, the addition rule states $P(A \text{ or } B) = P(A) + P(B)$. This rule does NOT apply when events can overlap (e.g., drawing a card that is both a King and a Heart). The addition rule helps decode gambling advertisements that imply bettors can simply sum their chances — but only works when the underlying events truly cannot co-occur.

How does this explain the phenomenon?

The experimental die-roll data showed $P(\text{rolling a 2 or a 5}) \approx P(\text{rolling a 2}) + P(\text{rolling a 5}) = 1/6 + 1/6 = 2/6 = 1/3$. The Gor Mahia match analysis showed $P(\text{Win or Draw}) = 1/3 + 1/3 = 2/3$ because WIN and DRAW are mutually exclusive for one match. However, a betting advertiser who highlights $2/3$ is hiding $P(\text{Lose}) = 1/3$ and exploiting the fact that across many independent matches, the gambler's losses accumulate — consistent with what Lessons 1–3 established about the independence of each match result.

LESSON 5 (40 minutes): Probability of Single-Chance Mutually Exclusive Events — The Addition Rule in Action

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the addition rule $P(A \text{ or } B) = P(A) + P(B)$ to calculate the probability of mutually exclusive events in a single chance experiment, using a bag of mandazi-flavour tokens as the anchoring experiment.
Knowledge	– Define mutually exclusive events as events that cannot occur at the same time on a single trial – State and apply the addition rule: $P(A \text{ or } B) = P(A) + P(B)$ for mutually exclusive events – Explain why the addition rule only works when events truly cannot co-occur – Recognise that the sum of probabilities of all mutually exclusive outcomes equals 1
Skills	– Carry out a single-draw experiment with coloured tokens and record outcomes systematically – Calculate $P(A \text{ or } B)$ using the addition rule across varied contexts (spinner, die, days of the week) – Peer-mark worked solutions against a mark scheme and give constructive feedback
Attitudes	– Show curiosity and openness when experimental results align with — or slightly deviate from — theoretical predictions – Appreciate that the addition rule exposes the mathematical flaw in gambling claims that imply probabilities can be freely combined to guarantee wins
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	Having established in Lesson 4 that mutually exclusive events cannot co-occur and that $P(A \text{ or } B) = P(A) + P(B)$, learners now put the addition rule to work through hands-on experimentation and multiple worked contexts — moving from identifying mutually exclusive events to confidently calculating combined probabilities, and directly challenging betting-advertisement logic that misapplies probability addition.
Safety Notes	No specific hazards. Ensure tokens/counters are not swallowed — remind learners these are mathematical tools, not food, even though they represent mandazi flavours.

B. LESSON OVERVIEW

This lesson is the computational payoff of the mutually exclusive events thread that began in Lesson 4. Learners move from identifying mutually exclusive events to fluently calculating combined single-chance probabilities using the addition rule $P(A \text{ or } B) = P(A) + P(B)$. The lesson is anchored in a tactile experiment: a cloth bag containing 15 coloured tokens representing three mandazi flavours sold at a school canteen — 6 plain (white), 5 coconut (yellow), and 4 cardamom (green). Each group draws one token, records the outcome, replaces it, and repeats 20 times. This mirrors a single-chance scenario: on any one draw, can you get plain AND coconut simultaneously? No — the events are mutually exclusive. This grounds the rule in physical reality before any algebra appears.

From the experiment, learners calculate $P(\text{plain})$, $P(\text{coconut})$, and $P(\text{plain OR coconut})$ both from their tally data and theoretically, comparing the two and discussing why slight differences arise. The teacher then leads three worked examples in escalating complexity: a spinner divided into unequal sectors (representing matatu route colours), a standard die where the event is rolling an even number OR a 5, and selecting a school day that is Monday OR Friday from the five-day week. For each, learners first attempt the problem independently for 90 seconds before the teacher reveals the solution — a structured think-before-see routine that prevents passive copying. Learners then complete a four-question independent practice set and peer-mark using a projected mark scheme.

The lesson closes by connecting the addition rule back to the driving question: gambling advertisements on Nairobi matatus and social media often imply that combining multiple bets multiplies your chances of winning. The addition rule reveals this is only mathematically valid when events are truly mutually exclusive AND when the probabilities are correctly assigned — two conditions that betting companies exploit through ambiguity. Updating the Driving Question Board and revising the cumulative class model cements the addition rule as the next permanent tool in learners' probability toolkit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Addition rule for probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mutually Exclusive Events (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner reads a short scenario on their ARES device or printed card: 'A school canteen sells three types of mandazi — plain, coconut, and cardamom. Amina reaches into a bag of 15 mandazi tokens without looking and pulls one out. What is the probability she gets a plain OR a coconut mandazi? Write your prediction and your reasoning before the experiment.' Learners write their prediction individually in their exercise books — including their working, not just a number — then share with a shoulder partner for 60 seconds.	Display the bag of tokens at the front — hold it up so learners can see it is opaque. Say: 'Before I tell you anything about what is inside, I want you to think about what we learned last lesson about the addition rule. Write your prediction now — you have 90 seconds, working alone.' [Start timer. Circulate silently. Do NOT answer questions about bag contents yet.] After 90 seconds: 'Share your prediction with your partner. You have 60 seconds — listen first, then respond.' After sharing, cold-call 3 learners with different predictions: 'Achieng, what did you predict and why?' [Wait 10 seconds.] 'Omondi, yours was different — walk us through your thinking.' [Wait 10 seconds.] 'Wanjiku, did anyone in your pair change their mind after talking? What shifted?' [Wait 12 seconds.] Do NOT confirm or correct yet. Write the three different predictions on the board under the heading 'Our Predictions — We Will Test These.'	Predict-Before-Observe (PBO): By committing a written prediction to paper before any data is collected, learners activate prior knowledge from Lessons 2–4 (probability formula, mutually exclusive events, addition rule) and surface misconceptions — particularly the tendency to add probabilities without checking mutual exclusivity. Visible predictions on the board create accountability and a concrete record to revisit.	Teacher scans written predictions during circulation. Key indicators: (1) Does the learner attempt to use $P(A \text{ or } B) = P(A) + P(B)$, suggesting retention from Lesson 4? (2) Does the learner correctly identify that plain and coconut are mutually exclusive on a single draw? (3) Are any learners simply guessing 50/50 or 'more than half' without calculation? Flag the third group for targeted questioning during the Observe phase.
Observe Phase  VIDEO: Mutually Exclusive Events Principles Source: CK-12  Search ARES	Groups of 4 receive a cloth bag containing 15 coloured tokens: 6 white (plain mandazi), 5 yellow (coconut mandazi), 4 green (cardamom mandazi). One learner draws a token, the recorder notes the colour on a tally chart, the	Before distributing bags, say: 'I am now going to tell you what is in the bag. There are 15 tokens: 6 white for plain mandazi, 5 yellow for coconut, and 4 green for cardamom. Each group will do 20 draws with replacement. Your job	Hands-On Experiment with Class Data Aggregation: The physical draw makes mutual exclusivity viscerally real — learners cannot pull a token that is simultaneously two flavours. Aggregating group results on a shared class grid	Teacher checks tally charts for systematic recording (not guessed totals). Key observable indicators: (1) Groups correctly divide tally frequency by 20 to get experimental probability — not by 15. (2) Groups sum $P(\text{plain}) +$

for similar videos EXERCISE: Mutually Exclusive Events Practice Source: CK-12 Search ARES for similar readings	token is replaced and the bag is shaken, and the next learner draws. Groups complete 20 draws. After tallying, groups calculate three experimental probabilities: P(plain), P(coconut), and P(plain OR coconut) from their own data. Groups record results on a shared class grid drawn on the board (or projected on ARES), allowing the whole class to compare across groups.	is to tally each colour and then calculate three probabilities from your data.' Distribute bags and tally sheets. [Allow 6 minutes for draws.] Circulate and prompt: 'Why do we replace the token each time? What would change if we did not replace it?' [Wait 10 seconds at each group.] After draws: 'Calculate your three probabilities now — P(plain), P(coconut), P(plain OR coconut) — from your tally. You have 2 minutes.' After groups post results on the class grid, ask: 'Look at the grid. Are all groups getting exactly the same answer for P(plain OR coconut)? Why or why not?' Cold-call 2 learners: 'Kipchoge, what do you notice?' [Wait 12 seconds.] 'Fatuma, what does that tell us about experimental versus theoretical probability?' [Wait 10 seconds.] Now ask the whole class: 'Can a single token be BOTH plain AND coconut at the same time?' [Wait 5 seconds for choral response.] 'So these events are — ?' [Elicit: mutually exclusive.] 'Now let us calculate what the theory says the probability SHOULD be.'	surfaces natural variation in experimental probability and motivates the need for theoretical calculation, bridging from empirical data to the formal addition rule. Connecting the tokens to mandazi — a familiar Kenyan snack — lowers affective barriers and keeps the phenomenon anchor (Gor Mahia, betting, chance) accessible.	P(coconut) from their tally correctly. (3) Any group reporting $P(\text{plain OR coconut}) > 1$ signals a calculation error — intervene immediately with: 'Can a probability ever be greater than 1? What does that tell you about your arithmetic?'
Explain Phase VIDEO: Addition rule for probability Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Mutually Exclusive Events (Flexbook) Source: CK-12 Search ARES	The teacher leads the class through the theoretical calculation for the mandazi bag, then works through three further examples on the board with increasing complexity — a spinner, a die, and days of the school week. For each example after the first, learners are given 90 seconds to attempt the problem independently in their books before the teacher reveals the solution. After all worked examples, learners complete a four-question independent practice set (displayed on ARES or the	WORKED EXAMPLE 1 — Mandazi Bag (whole class): Write on board: 'Bag: 6 plain, 5 coconut, 4 cardamom. Total = 15. One token drawn at random.' Ask: 'What is P(plain)?' [Cold-call: 'Muthoni?'] [Wait 10 seconds.] Write 6/15. 'P(coconut)?' [Cold-call: 'Otieno?'] [Wait 10 seconds.] Write 5/15. 'Are plain and coconut mutually exclusive on one draw?' [Choral: Yes.] 'So P(plain OR coconut) = ?' [Cold-call: 'Zawadi?'] [Wait 10 seconds.] Write $6/15 + 5/15 = 11/15$. 'Now compare this to	Gradual Release with Independent Attempt Before Reveal (I Do → You Try → We Confirm): The teacher models the first example fully, then requires learners to attempt each subsequent example alone before seeing the solution. This prevents passive transcription and forces learners to activate the addition rule from memory. Peer-marking with a mark scheme develops metacognitive awareness — learners must evaluate mathematical reasoning, not just circle right/wrong answers.	During independent practice, teacher circulates and targets three observable errors: (1) Learner adds $P(A) + P(B)$ without checking overlap — intervene with: 'Before you add, ask yourself: can both happen on the same trial?' (2) Learner uses total tokens as denominator inconsistently (e.g., uses 15 for one fraction and 20 for another) — redirect to: 'What is the total number of equally likely outcomes for one draw?' (3) Learner gets a probability greater than 1 — ask: 'Is that in the range

for similar readings	board), then peer-mark using a projected mark scheme.	your experimental result from your group. Are they close? Why might they differ slightly?' [Wait 12 seconds. Take 2 hands.] WORKED EXAMPLE 2 — Spinner (independent attempt first): Display: 'A spinner has 8 equal sectors: 3 red (Eastleigh route), 2 blue (Westlands route), 2 green (Karen route), 1 yellow (CBD route). One spin. Find P(red OR green).' Say: 'Attempt this yourself — 90 seconds, books only.' [Time.] Then cold-call: 'Amina, show us your working.' [Wait 10 seconds.] Confirm: $P(\text{red}) = 3/8$, $P(\text{green}) = 2/8$, $P(\text{red OR green}) = 5/8$. Ask: 'Could the spinner land on red AND green at the same time?' [Wait 5 seconds.] 'Good — that confirms mutual exclusivity.' WORKED EXAMPLE 3 — Die (independent attempt): Display: 'A fair die is rolled once. Find P(even number OR 5).' Say: '90 seconds independently.' [Time.] Cold-call: 'Baraka.' [Wait 10 seconds.] Walk through: even = {2,4,6} so $P(\text{even}) = 3/6$; 5 is odd so {5} does not overlap with even; $P(5) = 1/6$; $P(\text{even OR } 5) = 3/6 + 1/6 = 4/6 = 2/3$. Ask: 'Why must we check overlap before applying the addition rule? What would go wrong if we applied it to events that CAN co-occur?' [Wait 12 seconds. Cold-call 2 learners.] WORKED EXAMPLE 4 — Days of Week (independent attempt): Display: 'A school timetable coordinator randomly selects one day from the five school days (Monday–Friday) for a special assembly. Find P(Monday OR Friday).' '90 seconds.' [Time.] Cold-call: 'Nyambura.' [Wait 10 seconds.] Confirm: $P(\text{Monday}) = 1/5$, $P(\text{Friday}) = 1/5$, $P(\text{Monday OR$	The gambling connection embeds the KICD PCI (Life Skills: Decision Making) by showing the rule has real protective value.	we established in Lesson 3? What does that signal?' Peer-marking reveals systematic errors that teacher notes for Lesson 6.
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		Friday) = $2/5$. INDEPENDENT PRACTICE (4 questions — 5 minutes): Display on board/ARES. Q1: Bag has 4 red, 3 blue, 3 green beads. One draw. $P(\text{red OR blue}) = ?$ Q2: A spinner has sectors: 5 orange, 3 purple, 2 white. One spin. $P(\text{orange OR white}) = ?$ Q3: A card is drawn from cards numbered 1–10. $P(\text{prime OR } 10) = ?$ Q4: A football match can end in Win (W), Draw (D), or Loss (L) for Gor Mahia, with $P(W)=0.5$, $P(D)=0.25$, $P(L)=0.25$. Find $P(W \text{ OR } D)$. After 5 minutes: 'Swap books with your partner and use the mark scheme on the board to mark. Circle any step where your partner's working is unclear — not just the answer.' Project mark scheme. Debrief Q3 especially: 'Primes from 1–10 are 2,3,5,7 — that is 4 primes. $P(\text{prime}) = 4/10$. Is 10 a prime? No. So are prime and 10 mutually exclusive? Yes. $P(\text{prime OR } 10) = 5/10 = 1/2$.' GAMBLING CONNECTION: Ask: 'Betting apps often show odds for multiple outcomes and suggest you can combine them easily. What condition must be true before you can simply add probabilities? Are betting outcomes always mutually exclusive?' [Wait 15 seconds.] Cold-call: 'Kamau, what is the danger here?' [Wait 10 seconds.]		
Driving Question Board (DQB) Creation  VIDEO: Mutually Exclusive Events Principles Source: CK-12  Search ARES	Learners review the class Driving Question Board displayed at the front of the room. Working in pairs for 2 minutes, they discuss two prompts written on the board: (1) 'What new mathematical tool did we confirm today, and when can we use it?' (2) 'Does knowing the addition rule change what you think about the Gor Mahia betting	Direct learners to the DQB. Say: 'Look at where we started in Lesson 1 — Gor Mahia fans saying the team was due for a loss. Look at what we have added since. Today we confirmed a rule that lets us calculate combined probabilities. Discuss with your partner: what do we now know that we did not know before Lesson 5?	Driving Question Board Update (DQB Revision): The DQB functions as a living, visible record of the class's growing understanding. By revisiting it each lesson and adding only statements learners can now defend with evidence and calculation, it externalises conceptual growth and motivates the unresolved	Teacher evaluates the quality of DQB contributions: (1) Can learners articulate the addition rule as a conditional statement ('We can add probabilities IF the events are mutually exclusive')? (2) Are learners generating questions about the limits of the rule (e.g., 'What if events overlap?'), indicating readiness for

for similar videos  EXERCISE: Mutually Exclusive Events Practice Source: CK-12  Search ARES for similar readings	advertisements?' Each pair nominates one new statement or question to add to the DQB. The class votes (thumbs up) on the three strongest additions, which are physically added to the board.	And what new questions does this raise? Two minutes.' [Time.] Circulate and listen for: references to mutual exclusivity as a condition, questions about non-mutually-exclusive events (overlap), or questions about whether betting companies use the correct rule. After 2 minutes: 'I heard some excellent ideas. Who has a new statement for the board — something we can now say with confidence?' [Take 3–4 hands. Scribe statements.] 'Who has a new question — something today's lesson made you wonder about?' [Take 2–3 hands. Scribe questions.] Highlight any student-generated question about overlapping events (e.g., 'What if two events CAN happen at the same time?') and say: 'Hold that thought — it becomes critical in our next lessons on independent events.' Point to the driving question displayed at the top of the DQB and ask: 'Are we any closer to answering why understanding probability protects us from gambling? What would you say now that you could not say in Lesson 1?' [Wait 15 seconds. Cold-call: 'Zipporah.'] [Wait 10 seconds.]	questions that drive the next lesson. Connecting today's addition rule back to the original Gor Mahia/betting phenomenon maintains the storyline thread and prevents the mathematics from feeling abstract.	independent events in Lesson 6? (3) Does at least one learner connect the rule to the gambling/betting phenomenon unprompted? If not, the gambling connection in the Explain phase was not internalised and should be revisited at the start of Lesson 6.
Model Building Phase  VIDEO: Probability with Venn diagrams <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners retrieve their cumulative probability model — a diagram or written framework they have been building since Lesson 1, either in their exercise books or on a shared poster. They add today's learning as a new layer: a labelled box or branch for 'Mutually Exclusive Events — Single Chance' that includes the addition rule formula, the condition for using it, and one self-chosen example. Learners annotate their	Say: 'Take out your probability model — the diagram you have been building since Lesson 1. We are going to add one more layer today. You have 4 minutes. Add: the addition rule formula, the condition that must be met, one example you solved today, and an arrow back to the phenomenon with a sentence explaining how this rule helps someone think critically about a betting advertisement.' [Time 4 minutes.	Cumulative Model Revision (Layered Concept Map): The model is not redrawn each lesson but extended — each layer references and connects to prior layers, making conceptual dependencies visible. The mandatory annotation linking the addition rule to the betting/Gor Mahia phenomenon ensures every mathematical tool is contextualised within the driving question, preventing inert knowledge.	Teacher checks that each model contains: (1) The addition rule formula written correctly: $P(A \text{ or } B) = P(A) + P(B)$. (2) An explicit condition: 'only when A and B are mutually exclusive.' (3) A phenomenon-linked annotation that goes beyond 'gambling is bad' to include mathematical reasoning (e.g., 'Adding probabilities only works if the outcomes cannot both happen — betting ads often imply you can add chances that actually

 HTML: Probability of Compound Events (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	model with an arrow connecting the addition rule back to the Gor Mahia / betting phenomenon with a brief written note explaining why a bettor who blindly adds probabilities may be making a mathematical error.	Circulate.] Look specifically for: learners who write the formula without the condition ('only when mutually exclusive') — tap their book and ask: 'What must be true first?' [Wait 8 seconds.] After 4 minutes, ask 2 learners to share their phenomenon annotation. Cold-call: 'Njoroge, read me your sentence about betting.' [Wait 10 seconds.] 'Akinyi, does your annotation agree or add something different?' [Wait 10 seconds.] Close the lesson: 'By the end of Lesson 1, we had a question: if each match is independent, why do fans believe past results change future probabilities? We now have five tools in our model. In our next lessons, we will tackle independent events directly — and see what happens when we run two experiments, not one. Keep your model safe.'	Sharing annotations aloud creates a low-stakes accountability structure and allows the teacher to hear how learners are constructing the phenomenon explanation.	overlap, making the combined probability seem higher than it really is'). Absence of the condition in (2) is the highest-priority misconception to address at the start of Lesson 6.
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D. TEACHER REFLECTION

1. During the mandazi-bag experiment, did learners spontaneously identify mutual exclusivity from the physical activity, or did they only recognise it after I named it? What does this tell me about the depth of conceptual understanding from Lesson 4 — and how should I open Lesson 6?
2. In the independent practice, which of the four questions produced the most errors? Was Q3 (prime OR 10) — which required learners to first identify the set of primes — the sticking point, or was Q4 (Gor Mahia match outcomes with given probabilities rather than calculated ones) harder? How should I differentiate in Lesson 6 based on peer-marking data?
3. Did learners' DQB contributions show genuine connection between the addition rule and the gambling/betting phenomenon, or were the connections superficial (e.g., 'gambling is bad')? If superficial, which part of the Explain phase failed to land the mathematical critique of betting logic?
4. Did I give sufficient wait time (10–15 seconds) consistently, especially after higher-order questions like 'What is the danger of applying the addition rule without checking overlap?' Review the lesson mentally — were there moments I answered my own question before a learner could respond?
5. Are there learners whose cumulative models reveal a persistent gap — for example, confusing mutually exclusive events with equally likely events, or failing to apply the formula as a conditional rule? How will I provide targeted support for these learners before Lesson 6 introduces the two-event independent events chain?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We drew coloured tokens (representing plain, coconut, and cardamom mandazi) one at a time from a bag of 15, replaced each token after drawing, and recorded 20 draws per group. We saw that no single draw could produce two flavours simultaneously — the outcomes were mutually exclusive. Our experimental probabilities for P(plain OR coconut) were close to, but not exactly equal
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	to, the theoretical value of 11/15.
What did I learn?	When two events are mutually exclusive — meaning they cannot both occur on a single trial — the probability of either one occurring is found by adding their individual probabilities: $P(A \text{ or } B) = P(A) + P(B)$. This rule was confirmed across four different contexts: the mandazi bag, a matatu-route spinner, a standard die, and the school-week days. The rule only applies when events truly cannot co-occur; applying it blindly to overlapping events gives an incorrect (inflated) probability.
How does this explain the phenomenon?	Betting advertisements on Nairobi matatus and social media often imply that you can freely combine the probabilities of different outcomes to increase your chances of winning. Today's addition rule reveals this is only mathematically valid when the events are genuinely mutually exclusive and when each individual probability is correctly calculated. In a Gor Mahia match, Win, Draw, and Loss are mutually exclusive — but betting companies frequently present combinations of outcomes in ways that blur this condition, making combined probabilities appear more favourable than they truly are. This is one of the mathematical tricks that makes gambling dangerous.

LESSON 6 (40 minutes): Independent Events — The Coin Has No Memory

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners carry out two-stage experiments — tossing a coin AND rolling a die, then drawing a card with replacement — to derive and formalise the multiplication rule $P(A \text{ and } B) = P(A) \times P(B)$ for independent events, directly explaining why a coin cannot 'remember' its last result.
Knowledge	– Define independent events as events where the outcome of one trial has no effect on the outcome of another. – State and apply the multiplication rule: $P(A \text{ and } B) = P(A) \times P(B)$ for two independent events. – Construct a complete sample space for a two-stage experiment using a tree diagram and a two-way table.
Skills	– Carry out two-stage probability experiments (coin-and-die; card-draw-with-replacement) and record all outcomes systematically. – Draw tree diagrams and two-way tables to list sample spaces and calculate probabilities of combined outcomes. – Derive the multiplication rule from experimental frequency data and verify it against theoretical values.
Attitudes	– Appreciate that mathematical independence — not luck, streaks, or fate — governs repeated chance events, building resistance to gambling fallacies. – Demonstrate responsibility by making decisions about gambling risks using evidence from probability mathematics rather than intuition or peer pressure.
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	This lesson is the conceptual turning point of the sequence: having established equally likely outcomes (L1–L2), the range of probability (L3), and the addition rule for mutually exclusive events (L4–L5), learners now discover that when two events are independent, their joint probability is found by multiplying — and crucially, that independence means past outcomes carry zero information about future ones, mathematically dismantling the gambler's fallacy surrounding Gor Mahia's winning streak and the betting advertisements on Nairobi matatus.
Safety Notes	No specific hazards. Ensure coins and die are not swallowed or thrown. If using playing cards, handle respectfully and collect all cards after the activity.

B. LESSON OVERVIEW

This lesson is the pivotal sixth instalment in the eight-lesson evidence-gathering sequence on Probability I. Having spent five lessons building the conceptual foundations — equally likely events, theoretical probability, the range $0 \leq P \leq 1$, mutually exclusive events, and the addition rule — learners are now ready to tackle the most counterintuitive idea in the entire unit: that two events can be genuinely independent, meaning the result of the first gives absolutely no information about the result of the second. This is the mathematical reason a coin cannot be "due" for tails and Gor Mahia cannot be "due" for a loss.

The lesson is anchored in two concrete, hands-on experiments. First, learners simultaneously toss a coin AND roll a standard die — a direct two-stage experiment that mirrors the structure of the anchoring phenomenon (a football match result followed by another match result). They record all 12 possible outcomes, organise them into a tree diagram and a two-way table, compute experimental relative frequencies, and then compare these to the theoretical probabilities obtained by multiplying $P(\text{coin outcome}) \times P(\text{die outcome})$. The Kisumu rainfall context is woven in as a supporting phenomenon: whether it rains in Kisumu today has no effect on whether a coin lands heads — two events from completely different systems are independent. Second, learners draw a card from a shuffled set of 10 numbered cards, record the

outcome, replace the card, and draw again — reinforcing that replacement restores the original sample space and guarantees independence.

From these two experiments, learners derive the multiplication rule $P(A \text{ and } B) = P(A) \times P(B)$ as a pattern in their own data before the teacher formalises it. The rule is then connected explicitly back to the driving question: because each football match is an independent event — just like each coin toss — Gor Mahia's seven consecutive wins do not raise or lower the probability of the eighth match's outcome by even one decimal place. The lesson closes with an update to the Driving Question Board and a revision of each learner's growing probability model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Compound probability of independent events - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Probabilities of compound events Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown a scenario card on the board: 'It has rained in Kisumu for the past three days. Akinyi is about to toss a coin. Does the Kisumu rain make heads more likely, less likely, or equally likely as before?' Each learner writes a private prediction in their exercise book — including a one-sentence justification — before any discussion. Then learners are shown a second scenario: 'Gor Mahia has won their last seven Premier League matches. They play Tusker FC next Saturday. Is a Gor Mahia WIN now more likely, less likely, or the same probability as it was before the streak?' Learners write a second prediction and justification. Learners hold their predictions — they will test them experimentally.	Write both scenarios on the board before class begins. Open with: 'Before we touch any equipment today, I want your honest gut feeling — no right or wrong answer yet.' Distribute the scenario cards or point to the board. Say: 'Read Scenario 1. Write your prediction AND your reason in your book. You have 90 seconds — go.' Allow 90 seconds of silent individual writing. After time, say: 'Now read Scenario 2. Same thing — prediction and reason. 90 more seconds.' After writing, cold-call 4 learners (aim for a mix of genders and seating positions): 'Baraka, what did you predict for the rain-and-coin scenario, and why?' Wait 10 seconds after each response before calling the next. Do NOT confirm or correct any prediction — say only: 'Interesting. Hold that thought. Our experiment today will help us check.' Record on the board a tally of how many predicted MORE LIKELY / LESS LIKELY / SAME for each scenario — keep this visible throughout the lesson.	Predict-Before-Observe (Elicitation of Prior Conceptions): By committing a written prediction to paper before any instruction, learners activate their intuitive models of probability — including the gambler's fallacy if present. The Kisumu rain contrast (clearly a different system from a coin) is designed to create productive cognitive dissonance: most learners will correctly see that rain cannot affect a coin, but then must reckon with why a football streak feels different. This dissonance is the motivating tension the experiment will resolve.	Circulate during the 90-second writing windows and scan books for justifications. Look for: (a) learners who correctly predict SAME PROBABILITY for both scenarios and cite independence as a reason — they are ahead of the concept and can be asked to share their reasoning during the Explain Phase; (b) learners who predict MORE LIKELY or LESS LIKELY for the coin scenario — this flags a strong gambler's fallacy misconception that must be addressed explicitly in the Explain Phase; (c) learners who give different answers for the two scenarios (correct for coin, incorrect for football) — they sense independence in one context but not the other, which is the most common pattern and the richest teaching opportunity.
Observe Phase  VIDEO: Die rolling	Learners work in groups of four. Each group receives one coin, one standard die, and a pre-printed recording table with columns: Toss	Distribute materials and say: 'You are going to run two experiments today. In Experiment 1, you toss the coin and roll the die at the	Simultaneous Multi-Representation (Tree Diagram + Two-Way Table + Frequency Data): Learners build three parallel	During the tree diagram construction, look for: (a) groups that draw only one level of branches — they have not

probability with independent events - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos  HTML: Probability of Independent Events (Flexbook) Source: CK-12 Search ARES for similar readings	Number Coin Result (H/T) Die Result (1–6) Combined Outcome. Groups conduct 24 tosses of the coin-and-die together (simultaneously), recording every combined outcome. While one pair tosses and rolls, the other pair records. After 24 trials, each group lists all unique combined outcomes observed and any they did NOT observe, then constructs a tree diagram starting from the coin branch (H / T) and extending to six die branches under each, labelling all 12 possible combined outcomes (H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6). Groups also complete a two-way table with Coin (rows: H, T) against Die (columns: 1–6), placing a tick in each cell as outcomes occur. For the second experiment, groups draw from a shuffled pack of 10 numbered cards (1–10), record the number, replace and reshuffle, then draw again — repeating 20 times — recording the pair of draws each time. Learners note whether their second draw feels 'influenced' by the first.	same time — simultaneously. Record every single result. You have 10 minutes.' Set a visible timer. After 5 minutes, pause and circulate: check that groups are recording combined outcomes (not just coin or just die separately). If a group records them separately, say: 'We need to know BOTH results together — each row is one trial with a coin AND a die result.' After 10 minutes, say: 'Now draw your tree diagram. Start with two branches — Heads and Tails. Then from each branch, draw six more branches — one for each face of the die. Label every end branch with the combined outcome like H3 or T5.' Allow 5 minutes. Cold-call one group to display their tree diagram and explain one branch: 'Group from the back row — Nafisa, trace the path to outcome T4 on your tree and tell us what it means.' Wait 12 seconds. Transition to Experiment 2: 'Now the card draw. Replace the card every time. Why do you think I said replace? Write a one-sentence hypothesis.' Allow 30 seconds, then say: 'We will come back to that.' Monitor that groups shuffle between draws. After 8 minutes total for Experiment 2, call time.	representations of the same sample space. The tree diagram makes the sequential, branching structure of two-stage experiments visible. The two-way table organises outcomes spatially and makes it easy to count. The frequency tally from actual trials provides empirical data to compare against theoretical predictions. Using all three together prevents learners from treating any single representation as the 'real' answer and builds flexible, robust understanding of sample space.	grasped the two-stage structure; prompt with 'After the coin lands, the die is still rolled — what are the possible die results after Heads?'; (b) groups that list only outcomes they actually observed and omit unobserved ones — prompt with 'Your tree should show what COULD happen, not only what DID happen today'; (c) correct tree diagrams with all 12 endpoints labelled. For the two-way table, check that every cell is populated with either a tally or a zero. Ask one learner per group: 'How many different combined outcomes are possible?' — the correct answer is 12, confirming they have grasped the size of the sample space.
Explain Phase  VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE:	Groups use their tree diagrams to calculate the theoretical probability of each of the 12 combined outcomes. Learners first calculate $P(H) = 1/2$, $P(T) = 1/2$, and $P(\text{any specific die face}) = 1/6$, recording these as fractions. They then calculate $P(H) \times P(3) = 1/2 \times 1/6 = 1/12$ and compare this to the number of times 'H3' appears in their frequency table divided by 24. Groups discuss: 'What do you	Open the Explain Phase by writing on the board: $P(H) = \underline{\quad}$; $P(3) = \underline{\quad}$; $P(H) \times P(3) = \underline{\quad}$. Ask the whole class: 'Fill these in your book — 30 seconds.' After 30 seconds, cold-call a learner: 'Omondi, what is $P(H)$ times $P(3)$?' Wait 10 seconds. Write $1/12$ on the board. Then say: 'Now look at your frequency table. Out of 24 trials, how many times did H3 appear? Divide that by 24. Is it	Concept-from-Data Formalisation (Inductive Rule Derivation): Rather than presenting the multiplication rule as a given formula, learners see it emerge from their own experimental frequencies — the fraction of times H3 appeared in 24 trials approximates $1/12$, which equals $1/2 \times 1/6$. This inductive pathway gives the rule empirical grounding before it is stated abstractly. The strategy is	Monitor the three rapid practice problems. Correct answer benchmarks: (1) $P(T \text{ and } 5) = 1/12$; (2) $P(H \text{ and even number}) = 1/2 \times 3/6 = 1/4$; (3) $P(7 \text{ then } 7 \text{ with replacement}) = 1/10 \times 1/10 = 1/100$. Flag learners who add rather than multiply — this is the most common error and indicates confusion between the addition rule (mutually exclusive, 'or') and the multiplication rule

Probability of Independent Events Practice Source: CK-12  Search ARES for similar readings	notice about the theoretical value 1/12 and the fraction of times H3 appeared in our experiment?' The teacher then guides the whole class to formalise the pattern as $P(A \text{ and } B) = P(A) \times P(B)$, name it the multiplication rule for independent events, and define independence precisely. Learners copy the formal definition and rule into their books, then apply it to three rapid practice problems: (1) $P(T \text{ and } 5)$; (2) $P(H \text{ and an even number})$; (3) $P(\text{drawing a '7' on the first card draw and a '7' on the second card draw with replacement})$. Learners also revisit the two prediction scenarios from the Predict Phase and write a correction if needed.	close to 1/12? Raise your hand if your experimental fraction is between 1/24 and 3/24 — roughly around 1/12.' Count hands. Then say: 'Every group has one of the 12 cells — on a count of three, shout out your experimental fraction for your assigned outcome. Ready? 1, 2, 3.' Listen for approximate clustering around 1/12. Then say: 'So what rule are we seeing? If I multiply the individual probabilities, I get the probability of BOTH things happening together.' Write: $P(A \text{ and } B) = P(A) \times P(B)$. Say: 'This is the multiplication rule. But it only works when the events are INDEPENDENT. Independent means: knowing the result of A tells you nothing — absolutely nothing — about B. Does knowing the coin landed Heads tell you anything about what the die will show?' Wait 12 seconds. Cold-call: 'Wanjiku — yes or no, and why?' After the answer, say: 'Exactly. A coin and a die live in completely different worlds. They are independent.' Now connect to the phenomenon: 'Back to Gor Mahia. If each football match is an independent event — just like each coin toss — what does their five-win streak tell us about the probability of winning match six?' Wait 15 seconds. Cold-call 3 learners for brief responses. Close with: 'Write the formal definition of independent events and the multiplication rule in your book — leave space below to add an example in the next two minutes.' Give 2 minutes for the three rapid practice problems, then cold-call answers for each.	strengthened by the contrast with the addition rule from Lessons 4–5: learners must now distinguish 'A or B' (addition, mutually exclusive) from 'A and B' (multiplication, independent) — a critical conceptual boundary that prevents formula confusion in assessments.	(independent, 'and'). Ask those learners: 'Are we asking for A OR B happening, or A AND B both happening?' Also check that learners who revisit their Predict Phase predictions are writing a genuine correction where needed, not simply copying a peer's answer — ask them to explain the correction in their own words.
Driving	Each learner writes one new sticky	Say: 'Take one card from the pile	Sentence-Starter DQB Update	Read all posted cards before the

Question Board (DQB) Creation  VIDEO: Sample spaces for compound events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Independent Events and Sample Spaces (Flexbook) Source: CK-12  Search ARES for similar readings	note (or index card) to add to the class Driving Question Board. The note must complete ONE of the following sentence starters (displayed on the board): (A) 'I now understand that a coin cannot remember its last result because...'; (B) 'The multiplication rule $P(A \text{ and } B) = P(A) \times P(B)$ only works when...'; (C) 'This lesson changes my thinking about the Gor Mahia winning streak because...'; (D) 'Something I still wonder about independence is...'. Learners also update the 'What We Have Figured Out' column on the class DQB with a class-agreed statement: 'When two events are independent, knowing the result of the first event gives zero information about the result of the second. The probability of both occurring is found by multiplying their individual probabilities: $P(A \text{ and } B) = P(A) \times P(B)$.' Learners physically move or annotate any earlier DQB cards that are now answered by this lesson's evidence.	on my desk. Choose ONE of the four sentence starters on the board and complete it honestly — it can be a new understanding OR a new question. Two minutes, silent writing.' After 2 minutes, ask 6 learners to read their cards aloud (select 2 from each of the first three starters and at least 1 from starter D). After each reading, say: 'Which column does this belong in — what we figured out, or what we still wonder?' Let the learner place the card. Then point to the DQB's 'What We Have Figured Out' column and say: 'Who would like to suggest a class statement summarising what we proved today?' Cold-call one learner to draft it verbally; refine as a class in 2 exchanges; write the agreed statement on a large card and add it to the board. Finally, say: 'Look at the cards we posted in Lessons 1 through 5. Are there any questions on the board that today's experiment NOW answers? Point to one if you see it.' Give 30 seconds for learners to scan the board.	(Structured Reflection with Public Accountability): The four sentence starters are carefully designed to address different aspects of the lesson's conceptual load — the coin-memory misconception, the conditions for the multiplication rule, the phenomenon connection, and open-ended uncertainty. By requiring learners to choose and complete a starter publicly, the DQB becomes a living record of the class's growing understanding rather than a static display. Cards from Type D ('still wonder') are especially valuable as they seed the inquiry for Lessons 7 and 8.	next lesson. Cards for starters A and C should reference independence and the absence of memory/information transfer — vague answers like 'because it is random' are insufficient and indicate the learner has not yet grasped the mechanistic definition of independence. Cards for starter B should specify 'when the outcome of one event does not affect the other' — not simply 'when they are different events.' Collect any starter D cards that raise genuine new questions (e.g., 'What if the events are NOT independent — how does that change things?') and note them as potential launch points for Lesson 7.
Model Building Phase  VIDEO: Count outcomes using tree diagram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Independent Events and Sample Spaces Practice	Each learner returns to the probability model they have been building since Lesson 1 — a diagram or written explanation that tracks how probability works in the context of the anchoring phenomenon. Today, learners add a new labelled section titled 'Independent Events' and draw or annotate to show: (1) a two-branch tree for a two-stage experiment (e.g., Gor Mahia Match 1 outcome and Match 2 outcome); (2) the multiplication rule written symbolically and in words; (3) an arrow or annotation explicitly connecting the model to the driving	Distribute learners' model portfolios (or direct them to their model pages in their exercise books). Say: 'Open to your probability model. Today we are adding the most important layer yet. Label a new section: Independent Events.' Display the three required model elements on the board: (1) tree diagram for two-stage experiment, (2) multiplication rule in symbols and words, (3) one sentence linking independence to the driving question about Gor Mahia and gambling. Say: 'You have 5 minutes. Make it clear enough that a Form 3 student who	Cumulative Model Revision (Progressive Model Building): The model each learner is constructing across all eight lessons serves as a personalised, cumulative explanation of the phenomenon. By adding independence as a labelled layer today — with symbolic, verbal, and diagrammatic components — learners are forced to integrate the new concept with prior model elements (equally likely outcomes, probability range, mutually exclusive events). The requirement to include a sentence linking to the driving question	Scan models for three quality indicators: (a) the tree diagram must show two full stages with all branches labelled — a single-stage tree indicates incomplete understanding of two-stage experiments; (b) the multiplication rule must appear as both $P(A \text{ and } B) = P(A) \times P(B)$ and in plain language ('multiply the individual probabilities when events are independent'); (c) the driving-question sentence must use the word 'independent' and reference the absence of memory or predictive power — a sentence like 'because it is random' is

Source: CK-12 Search ARES for similar readings	question — explaining in one sentence why each match being an independent event means a winning streak carries no predictive power for the next match. Learners who finish early are invited to add a second example using the Kisumu rainfall context: $P(\text{rain today in Kisumu AND tails on a coin toss}) = P(\text{rain}) \times P(\text{tails})$.	missed today's lesson could understand your model.' Circulate and provide targeted feedback — do not correct models for learners, but ask probing questions: 'Your tree diagram shows the outcomes — now show me WHERE the $1/12$ comes from on your diagram.' With 90 seconds remaining, say: 'If you have finished the required three elements, add the Kisumu rain example as a bonus.' In the final 30 seconds, ask one learner to hold up their model and narrate it to the class in two sentences. Close the lesson by pointing to the DQ written on the board and saying: 'Our driving question asks why past outcomes do not change future probabilities. What is the one-word mathematical answer we discovered today?' Wait 10 seconds. Accept 'independence' — write it on the board next to the driving question in a different colour.	ensures that model building is not a mechanical diagram task but a genuine sensemaking act. The public narration at the close provides a final oral rehearsal of the key idea.	insufficient. Collect models at the end of class and review overnight; use findings to inform the opening of Lesson 7, which will extend these ideas to contexts with more complex sample spaces.
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D. TEACHER REFLECTION

1. During the Predict Phase, how many learners predicted the SAME PROBABILITY for both scenarios (rain-and-coin AND the football streak)? Did the learners who got both correct use the language of independence, or did they use intuition? How will I build on — or challenge — both groups in Lesson 7?
2. In the Observe Phase, did all groups successfully construct complete tree diagrams with all 12 endpoints labelled before the timer ended? If groups struggled, was the difficulty conceptual (not understanding two-stage structure) or procedural (not knowing how to draw a tree)? What scaffold would help in future lessons?
3. During the Explain Phase, how many learners added rather than multiplied when attempting the rapid practice problems? Is this confusion between $P(A \text{ or } B)$ and $P(A \text{ and } B)$ — i.e., between the addition rule from Lessons 4–5 and the new multiplication rule? What explicit contrast activity could I build into the opening of Lesson 7 to reinforce the boundary?
4. Looking at the DQB cards posted today — especially the 'still wonder' cards (starter D) — what genuine mathematical questions did learners raise? Are there questions about non-independent events, or about what happens when replacement is NOT used? How do these seeds connect to the content of Lessons 7 and 8?
5. Were learners able to articulate in their models — in their own words — why independence means that Gor Mahia's winning streak carries no mathematical information about the next match? Or did their model sentences remain vague? What additional questioning strategy could deepen this phenomenological connection

before the unit concludes?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners tossed a coin and rolled a die simultaneously for 24 trials, recording all combined outcomes and organising them into tree diagrams and two-way tables. They found approximately 12 equally likely combined outcomes, each occurring roughly 1/12 of the time. They also drew numbered cards with replacement and noticed the second draw was unaffected by the first.
What did I learn?	When two events are independent — meaning the outcome of one gives no information about the other — the probability of both occurring together is found by multiplying their individual probabilities: $P(A \text{ and } B) = P(A) \times P(B)$. Independence means the first event has no memory and cannot influence the second, regardless of what that first outcome was.
How does this explain the phenomenon?	Each Gor Mahia football match is an independent event — just like each coin toss and each die roll. The five-win streak carries zero mathematical information about the probability of winning match six. The multiplication rule quantifies joint probability without any adjustment for past results, proving that the gambler's fallacy promoted in betting advertisements on Nairobi matatus is mathematically false: no streak, however long, changes the probability of the next independent event.

LESSON 7 (80 minutes): Probability of Independent Events in Context: The Multiplication Rule

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the multiplication rule for independent events to multi-experiment, two-outcome scenarios drawn from Kenyan daily life — rainfall in Kisumu, school demographics, and a mandazi vendor's consecutive market-day sales — and use probability evidence to critically evaluate gambling claims.
Knowledge	Learners will know that: (i) two events are independent when the outcome of one does not affect the probability of the other; (ii) the probability of two independent events both occurring is found by multiplying their individual probabilities, $P(A \text{ and } B) = P(A) \times P(B)$; (iii) the multiplication rule extends the range-of-probability concept to compound events; (iv) the mathematical structure of independent events explains why a betting stall always profits in the long run.
Skills	Learners will be able to: (i) identify whether two events in a given scenario are independent; (ii) correctly apply the multiplication rule $P(A \text{ and } B) = P(A) \times P(B)$ to two-experiment, two-outcome problems; (iii) calculate and interpret compound probabilities in real-life Kenyan contexts; (iv) construct and use a probability tree diagram to organise all outcomes of two independent experiments; (v) use probability calculations to construct a reasoned argument against the gambler's fallacy and gambling culture.
Attitudes	Learners will: (i) appreciate that mathematics provides an objective tool for evaluating everyday claims about chance; (ii) demonstrate responsibility by applying probability knowledge to protect themselves and peers from the harms of gambling; (iii) show confidence in using the multiplication rule across unfamiliar contexts; (iv) value collaborative reasoning when checking each other's probability calculations.
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	In Lesson 6 learners established mutually exclusive events and single-experiment probabilities. Lesson 7 now extends the investigation to two-experiment scenarios, directly confronting the core phenomenon: if Gor Mahia's match outcomes are independent, can we calculate the exact probability of two consecutive wins — or two consecutive losses — using multiplication? Learners also examine the mandazi vendor and spinner/stall sub-phenomenon to prove mathematically that the 'house' always wins, dismantling betting-culture claims encountered on Nairobi matatus.
Safety Notes	No physical hazards. Ensure that the discussion of gambling dangers is handled sensitively — some learners may have family members affected by gambling addiction. Frame the conversation around mathematical evidence and financial protection, not personal judgment. Online safety: if learners use any digital device to access probability simulations, remind them to use only ARES-approved resources and not to share personal information.

B. LESSON OVERVIEW

Learners open by revisiting the Gor Mahia phenomenon and predicting the probability that the team wins two matches in a row. Through structured calculation tasks rooted in Kenyan contexts — Kisumu rainfall, Grade 10 school demographics, and a Nairobi mandazi vendor — they discover and formalise the multiplication rule for independent events. A probability tree diagram is used as the core sensemaking tool. Learners then analyse a spinner/betting-stall scenario to calculate why the stall profits over many trials, directly linking mathematics to the dangers of gambling. The DQB is substantially updated with new mathematical claims and revised models.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Compound probability of independent events - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Probability of Independent Events (Flexbook) Source: CK-12  Search ARES for similar readings	Learners are presented with three written scenario cards (distributed or displayed on screen): Card A — 'Weather in Kisumu: the probability it rains on any given day in April is 0.7. What is the probability it rains on BOTH Monday AND Tuesday?'; Card B — 'In a Nairobi school, 60% of Grade 10 learners are female. If you pick two learners at random and independently, what is the probability BOTH are female?'; Card C — 'A mandazi vendor in Gikomba market sells out her stock with probability 0.8 on any market day. What is the probability she sells out on TWO consecutive market days?' Each learner silently records a numerical prediction for all three scenarios in their exercise book, showing their reasoning. They then share predictions with a partner (Think-Pair), comparing methods and noting disagreements.	Distribute or project the three scenario cards. Say: 'Before we calculate anything, I want your best mathematical guess. Write a number between 0 and 1 for each scenario AND write at least one sentence explaining how you got it. You have 4 minutes — silent, individual work.' Set a visible timer. After 4 minutes, say: 'Now turn to your partner. Compare your answers. If you disagree, explain your reasoning to each other — do not just change your answer.' Allow 3 minutes for paired discussion. Cold-call 3 pairs (one per scenario) — choose learners with visibly different predictions: 'Pair near the window — what did you predict for Card A and how did you reason?' Apply 12-second wait time before accepting responses. Record 3–4 different predictions on the board for each scenario without commenting on correctness. Ask: 'Hands up if you multiplied the two probabilities together for any of these. Hands up if you added them. Hands up if you did something else entirely.' Tally responses visibly. Say: 'By the end of this lesson you will know exactly which operation to use and why — and you will be able to check whether your prediction was correct.'	Think-Pair-Share with prediction tallying. This surfaces prior conceptions (addition vs. multiplication, intuitive vs. systematic reasoning) and creates cognitive dissonance — multiple methods for the same problem. The public tally of operations (add/multiply/other) gives the teacher real-time data on the distribution of misconceptions across the class and sets up a concrete intellectual puzzle learners are motivated to resolve.	Teacher circulates during silent prediction time and scans exercise books. Observable indicators of understanding: (i) learner writes a value strictly between 0 and 1 (not 'impossible' or 'certain'); (ii) learner's reasoning references both individual probabilities, not just one; (iii) learner shows some calculation attempt rather than a guess. Red flag: learner adds the probabilities (e.g., $0.7 + 0.7 = 1.4$) — note this learner for targeted questioning in Phase 2.
Observe Phase  VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum)	Learners work in groups of four to carry out two structured activities. Activity 1 — Coin & Spinner Simulation (8 min): Each group receives (or draws) a simple two-section spinner labelled 'WIN' (probability 0.5) and 'LOSE' (probability 0.5), representing a simplified Gor Mahia match.	Introduce Activity 1: 'Each group — one person tosses the coin for Match 1, a different person tosses for Match 2. Record every trial in your table. You have 8 minutes.' Circulate and prompt: 'How many WIN–WIN results so far? What fraction of 20 is that?' After Activity 1, bring the class together for 2	Simulation-to-structure bridging with tree diagram construction. The coin simulation provides empirical data that learners can compare against a theoretical model they then build themselves. The tree diagram is the core representational tool: it makes the sample space of two independent	Observable indicators: (i) group's WIN–WIN experimental probability is in the range 0.20–0.30 (expected ~0.25); (ii) tree diagram has exactly four endpoints labelled with correct probabilities (0.49, 0.21, 0.21, 0.09 for the rain tree); (iii) endpoint probabilities sum to 1.00. Red flags: a group writes

Search ARES for similar videos EXERCISE: Probability of Independent Events Practice Source: CK-12 Search ARES for similar readings	Groups toss a coin twice (or spin twice) 20 times and record results in a two-column table (Match 1 result Match 2 result). They tally how many times 'WIN-WIN' occurred and calculate the experimental probability. Activity 2 — Tree Diagram Construction (12 min): Learners are guided through constructing a fully labelled probability tree diagram for two independent coin tosses, showing all four branches (HH, HT, TH, TT) with probabilities written on each branch and at each endpoint. They then redraw the tree for Card A (Kisumu rain: $P(\text{rain})=0.7$, $P(\text{no rain})=0.3$) and record all four endpoint probabilities, verifying they sum to 1.	minutes: 'Groups, call out your WIN-WIN experimental probability.' Record values on board. Say: 'They are all different — but they are all close to what value? Keep that number in mind.' Transition to Activity 2. Model the first branch of the rain tree diagram on the board: 'Monday: rain (0.7) or no rain (0.3). I draw two branches. On the rain branch — what happens on Tuesday?' Cold-call 2 learners using names: '[Name], what are the two possibilities for Tuesday if it rained Monday?' Apply 10-second wait time. Complete the tree collaboratively for the rain scenario. Then say: 'In your groups, verify that your four endpoint probabilities add to exactly 1. If they don't, find the error before moving on.' Give 4 minutes for group verification. Cold-call one group to explain their check: 'Group by the door — walk us through how you verified your tree summed to 1.'	experiments visible, shows why probabilities are multiplied (each branch probability is conditional on the path taken), and creates a reusable template for all three scenario cards. Verifying that endpoint probabilities sum to 1 builds self-checking habits.	$P(\text{rain on Tuesday} \text{it rained Monday}) = 0.7$ but writes a different value — suggests confusion about independence; a group's four endpoints sum to more or less than 1 — suggests branch probability errors. Teacher checks at least three groups' tree diagrams during the activity.
Explain Phase VIDEO: Compound probability of independent events - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos HTML: Probability of Independent Events (Flexbook) Source: CK-12 Search ARES for similar	Learners formalise the multiplication rule from their tree diagrams, then apply it to all three scenario cards and to the gambling/stall analysis. Step 1 — Formalisation (5 min): Guided by the teacher, learners write the multiplication rule in their books: 'For two independent events A and B, $P(A \text{ and } B) = P(A) \times P(B)$.' They verify this against their tree diagram endpoint for 'rain on both days': $0.7 \times 0.7 = 0.49$. Step 2 — Trio of Kenyan Contexts (12 min): In pairs, learners complete a structured worksheet solving all three scenario cards using the multiplication rule, drawing a tree diagram for each, and writing a	Step 1 — Write on the board: '$P(A \text{ and } B) = P(A) \times P(B)$ — The Multiplication Rule for Independent Events.' Say: 'Look at your tree diagram. The probability at each endpoint is found by multiplying along the branches. This is not a coincidence — it is always true for independent events. Write this rule in a box in your book — it is one of the most important results in this unit.' Step 2 — Distribute the worksheet. Say: 'Work in pairs. For each scenario card, you must: (1) draw the tree diagram, (2) state $P(A)$ and $P(B)$, (3) calculate $P(A \text{ and } B) = P(A) \times P(B)$, (4) write one sentence interpreting the answer for a real person who does not	Worked-example-to-application scaffolding with contextual interpretation. Learners move from abstract rule to three concrete Kenyan contexts and finally to a gambling-analysis problem. The requirement to write a plain-language sentence for each answer forces learners to connect symbolic mathematics to real-world meaning — a critical sensemaking move. The gambling stall calculation uses expected value reasoning (without formally naming it) to produce a striking numerical result (stall profits Ksh 1,800) that makes the abstract danger of gambling concrete and memorable, directly answering the	Observable indicators during pair work: (i) learner correctly identifies $P(\text{female}) = 0.6$ and applies $P(\text{both female}) = 0.6 \times 0.6 = 0.36$; (ii) learner's mandazi answer is $0.8 \times 0.8 = 0.64$; (iii) learner's plain-language sentence accurately reflects the numerical answer without rounding errors larger than 0.01. For the stall scenario: (i) $P(\text{win both}) = 1/3 \times 1/3 = 1/9 \approx 0.111$; (ii) stall profit calculation correct at Ksh 1,800. Red flag: learner writes 'the stall might profit or might lose' — indicates confusion between theoretical probability and outcome certainty; redirect by asking 'Over 180 spins, what does the mathematics predict

readings	one-sentence interpretation of each answer in plain language (e.g., 'There is a 49% chance it will rain in Kisumu on both Monday and Tuesday in April'). Step 3 — Mandazi Stall & Gambling Analysis (8 min): Learners read a short scenario: 'A betting stall near Ambassador Hotel, Nairobi, runs a spinner game. The spinner has 3 sectors: WIN (probability 1/3) and LOSE (probability 2/3). A player pays Ksh 20 to play. If they win, they receive Ksh 30 back (a profit of Ksh 10). If they lose, they lose their Ksh 20. Calculate: (a) P(player wins on a single spin); (b) P(player wins on BOTH of two consecutive spins); (c) If 90 learners each play twice, how much money does the stall expect to collect versus pay out?' Learners calculate individually, then compare answers with a partner.	study mathematics.' Circulate. At the 6-minute mark, cold-call 3 different pairs (one per scenario card) to share answers: '[Name], what did you get for the mandazi vendor? What does that number mean in real life?' Apply 12-second wait time. Correct misconceptions immediately: if a learner added probabilities, ask 'Does what happened on Monday change what happens on Tuesday for the vendor? No — so these are independent. Independent means we MULTIPLY.' Step 3 — Read the stall scenario aloud. Say: 'Calculate parts (a), (b) and (c) individually — 4 minutes.' After 4 minutes, cold-call: '[Name], what is the probability a player wins both spins?' Then: '[Name], if 90 learners each play twice — that is 180 spins — how much does the stall collect? How much does it pay out? What is the stall's profit?' Work through part (c) on the board with learner input: Collect = 180×20 = Ksh 3,600; Expected wins = $180 \times (1/3)$ = 60 spins; Payout = 60×30 = Ksh 1,800; Stall profit = Ksh 1,800. Say: 'The stall keeps Ksh 1,800 from 90 learners. This is not luck — it is mathematics. This is why gambling is dangerous.' Record this key statement on the DQB anchor strip.	driving question.	ON AVERAGE?'
Driving Question Board (DQB) Creation  VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum)	Learners individually complete three DQB sticky notes (or written entries in their DQB section of the exercise book): (1) a CLAIM note — one new mathematical claim they can now make about independent events, supported by a calculation from today's lesson; (2) a REVISION note — one idea or prediction from an earlier lesson that today's evidence has changed	Say: 'We have one lesson left in this sequence. Before we get there, we need to update our Driving Question Board to show how our thinking has grown. You have 5 minutes to write your three notes — CLAIM, REVISION, QUESTION — individually. Be specific. Do not write 'I learned about probability.' Write a mathematical statement.' Circulate	Claim-Evidence-Reasoning (CER) DQB update. The three-note structure (Claim, Revision, Question) requires learners to distinguish between what they know, what has changed, and what remains uncertain — a key epistemic sensemaking practice. The class carousel makes individual thinking public and allows the teacher to identify	Observable indicators: (i) CLAIM note contains a correct mathematical statement with a numerical example from today (not a restatement of the rule in isolation); (ii) REVISION note references a specific prediction or claim from a previous lesson (e.g., 'In Lesson 5 I thought past coin tosses change future ones — now I know P stays the same AND two

Search ARES for similar videos EXERCISE: Probability of Independent Events Practice Source: CK-12 Search ARES for similar readings	or confirmed, with an explanation; (3) a QUESTION note — one new question today's lesson has raised that they still want to investigate (e.g., 'What if the events are NOT independent — does the rule change?', 'Does the stall always profit even if it has a streak of losses?'). Groups then share their CLAIM notes with the class in a 3-minute carousel, and the teacher adds three key class-agreed claims to the DQB anchor board under the heading 'Lesson 7 — Independent Events.'	and prompt struggling learners: 'What can you NOW calculate that you could not calculate at the start of today?' After 5 minutes, facilitate the carousel: 'One person from each group — read your CLAIM note aloud in 15 seconds.' After all groups share, say: 'I am going to add three claims that came up repeatedly to our DQB.' Write: (1) '$P(A \text{ and } B) = P(A) \times P(B)$ for independent events'; (2) 'The probability of two independent events both occurring is LESS than either individual probability'; (3) 'Betting stalls are mathematically designed to profit — probability proves it.' Point to the driving question: 'Which part of our driving question can we now answer most completely?' Cold-call 2 learners. Apply 10-second wait time.	consensus claims versus persistent misconceptions. Anchoring new claims to the driving question maintains narrative coherence across the 8-lesson sequence.	independent flips multiply'); (iii) QUESTION note is genuinely new and mathematically meaningful, not restating today's content. Red flag: learner's CLAIM note is 'independent events do not affect each other' with no mathematical expression — prompt: 'Can you write that as a formula with numbers?'
Model Building Phase VIDEO: General multiplication rule example: independent events <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Probability with general multiplication rule <i>Related topic: not an exact match for this lesson.</i> Source: Khan	Learners retrieve their cumulative probability model (begun in Lesson 1 and revised in every lesson since). Today they add a new layer to the model: a two-experiment section showing a labelled probability tree diagram for two independent events, with the multiplication rule written as a key alongside it. They annotate the model with: (a) a Kenyan context example (their choice from today's three scenario cards); (b) a statement connecting the model to the driving question — specifically, how the multiplication rule explains why the gambler's fallacy is mathematically wrong when events are independent; (c) a colour-coded arrow linking back to the 'range of probability' and 'equally likely events' sections from earlier lessons, showing how independent-event probability still	Say: 'Open your cumulative probability model. You are going to add the most powerful tool we have built so far. Draw a probability tree diagram for two independent events — use any of today's three Kenyan scenarios as your example. Label every branch and endpoint. Write $P(A \text{ and } B) = P(A) \times P(B)$ as the KEY rule for this section.' Give 6 minutes for model revision. Circulate and prompt: 'Where in your model does this connect to what you drew in Lesson 1 about equally likely events?' and 'Can you show with your model exactly why someone who thinks the coin is DUE for tails is wrong?' After 6 minutes, cold-call one learner to hold up their model and describe one new feature they added today: '[Name], show us your tree diagram and explain in two sentences what it	Cumulative model revision with cross-lesson annotation. Adding the multiplication rule and tree diagram to the existing model forces learners to integrate new knowledge with prior learning (range of probability, equally likely events, mutually exclusive events) rather than treating each lesson as isolated content. The explicit annotation task — connecting the model to the driving question and the gambler's fallacy — requires learners to use the model as an explanatory tool, not merely a record. This mirrors how scientists use models: as living representations that grow with evidence.	Observable indicators: (i) tree diagram has correct branch probabilities and endpoint products; (ii) multiplication rule is written as a formula (not just in words); (iii) annotation explicitly names 'independent events' and references the driving question or Gor Mahia phenomenon; (iv) colour-coded arrow to earlier model sections is present and labelled. Exit check: as learners pack up, teacher asks 3 learners at random — 'Give me one number from today's lesson and tell me what it means.' Acceptable responses include: '0.49 — the probability of rain in Kisumu on both Monday and Tuesday'; '0.64 — the probability the mandazi vendor sells out two days running'; '1/9 — the probability a gambler wins both spins of the betting stall.' If a learner cannot produce a

Academy (English - US curriculum) Search ARES for similar readings	obeys $0 \leq P \leq 1$. Learners who finish early add a third experiment branch to their tree to explore what happens to compound probability as more independent events are chained.	proves about independent events.' Close the lesson: 'In our final lesson we will bring ALL of this together — equally likely events, mutually exclusive events, and independent events — to answer our driving question completely. Make sure your model is ready.' Preview: 'Lesson 8 will ask: if we know all of this mathematics, why do so many people still gamble? What does probability tell us about risk over time?'		number and meaning, note for follow-up at the start of Lesson 8.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Predict Phase, what proportion of learners chose to add the probabilities rather than multiply? Did the cold-call pairs help surface the range of misconceptions, or did you need to draw out more voices? (2) Were the three Kenyan scenario cards sufficiently differentiated — did the rainfall and mandazi contexts feel authentic to learners, or did any group struggle to connect to the context? Consider whether alternative contexts (e.g., probability that two Rift Valley maize farmers both have a good harvest in the same season) might resonate better with your specific class. (3) Did the gambling stall calculation produce the intended 'aha' moment? If learners seemed unmoved by the Ksh 1,800 profit figure, consider whether a more locally resonant monetary amount or a peer-narrated story would strengthen the emotional impact. (4) Were all learners able to complete a full probability tree diagram by the end of the Observe Phase? If not, identify whether the gap is procedural (drawing the tree) or conceptual (knowing what probabilities to place on each branch) — these require different interventions. (5) Review the DQB QUESTION notes: do learners' new questions point naturally toward Lesson 8 content (cumulative probability, risk over time, synthesising the full driving question)? If many questions are still about basic mechanics, plan a 5-minute consolidation opener for Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners tossed a coin (or used a spinner) twice in 20 trials and recorded WIN–WIN outcomes, finding an experimental probability close to 0.25. They constructed probability tree diagrams for two independent coin tosses and for the Kisumu rain scenario ($P(\text{rain})=0.7$), labelling all four branches and verifying that endpoint probabilities summed to 1.
What did I learn?	Two events are independent when the outcome of one does not affect the other. The probability of both independent events occurring is found by multiplying their individual probabilities: $P(A \text{ and } B) = P(A) \times P(B)$. This rule was applied to three Kenyan contexts: rain in Kisumu ($0.7 \times 0.7 = 0.49$), two female learners selected ($0.6 \times 0.6 = 0.36$), and a mandazi vendor selling out twice ($0.8 \times 0.8 = 0.64$). The compound probability of two independent events is always less than either individual probability. A betting stall running 180 spins at $P(\text{win})=1/3$ expects to pay out Ksh 1,800 while collecting Ksh 3,600 — a guaranteed mathematical profit of Ksh 1,800 — proving that gambling is structured to disadvantage the player.
How does this explain the phenomenon?	Because each match Gor Mahia plays is an independent event, the probability of the team winning two consecutive matches is $P(\text{win}) \times P(\text{win})$ — not a different probability influenced by their previous results. The gambler's fallacy claims that past outcomes change future probabilities, but the multiplication rule proves this is false: each event's probability is fixed, and the compound probability of a streak is simply the product of the fixed individual probabilities. This same mathematics explains why betting stalls always profit in the long run — the game is designed so that the house's expected payout is always less than its expected

	collection, regardless of short-term streaks.
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LESSON 8 (80 minutes): Consolidation, Final Explanation & DQB Closure: Why Mathematics — Not Luck — Governs the Odds

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all probability concepts — equally likely events, range of probability, mutually exclusive events, and independent events — and apply them to construct a complete, mathematically grounded explanation of the anchoring phenomenon, demonstrating why the gambler's fallacy is false and why gambling is a losing proposition.
Knowledge	Learners know that: (a) equally likely events have equal probability; (b) probability of any event lies in the range $0 \leq P(E) \leq 1$; (c) mutually exclusive events cannot occur simultaneously; (d) independent events have no influence on each other's probability; (e) for two independent events A and B, $P(A \text{ and } B) = P(A) \times P(B)$.
Skills	Learners are able to: perform and interpret experiments involving equally likely, mutually exclusive, and independent events; compute probabilities of single and compound independent events; construct a written and visual explanation connecting all four probability concepts to a real-world phenomenon; evaluate gambling claims using mathematical reasoning.
Attitudes	Learners appreciate the role of probability in making informed, responsible real-life decisions; demonstrate responsibility by applying probability knowledge to reject gambling fallacies; show integrity when presenting peer-reviewed mathematical arguments.
Key Inquiry Question	How do we apply probability in day-to-day life?
Purpose in Storyline	This is the culminating lesson of the sub-strand storyline. Learners have spent seven lessons building understanding of equally likely events, the range of probability, mutually exclusive events, and independent events — all anchored to the Gor Mahia FC winning-streak phenomenon and betting culture on Nairobi matatus. In this lesson, learners synthesise every concept into a final, complete explanation: each football match is an independent event, past results cannot change future probabilities (the gambler's fallacy is mathematically false), and gambling is designed so that $P(\text{bettor wins big})$ approaches 0 over repeated trials. Groups present mini-posters, the Driving Question Board is formally closed, a written assessment covers all SLOs, and learners write a personal reflection on one real-life decision they will make differently using probability knowledge.
Safety Notes	Discussions about gambling can be sensitive for learners whose families have been affected. Frame all references to gambling strictly as a mathematical and life-skills issue. Do not ask learners to share personal financial experiences publicly. Remind learners that the purpose of the lesson is empowerment through mathematics, not judgment of any individual or family.

B. LESSON OVERVIEW

Lesson 8 is the final, consolidation lesson for Sub-Strand 3.2: Probability I. It opens with a brief return to the anchoring phenomenon — clips of Gor Mahia FC winning-streak commentary and a mock betting-app advertisement displayed on the board — to re-anchor all prior learning. In the Predict Phase, learners individually write their current best answer to the Driving Question before any group work, surfacing growth in understanding. In the Observe Phase, groups receive a structured 'Evidence Pack' containing four probability problems drawn from all four concept areas (equally likely events, range of probability, mutually exclusive events, independent events), all set in Kenyan contexts (coin tosses, a Kisumu dice game, a Nairobi lottery scratch-card, and a Gor Mahia vs AFC Leopards match sequence). In the Explain Phase, groups construct a mini-poster that mathematically answers the Driving Question, connecting all four concepts. Groups present (gallery-walk format); peers use a structured feedback card. In the DQB Closure Phase, the class reviews every question posted across the eight lessons, marks each as 'Answered — here's the math', and identifies any remaining wonderings. In the Model Building Phase, learners compare their Lesson 1 initial model with a revised final model, write a personal reflection, and complete a short written assessment covering all SLOs. The lesson closes with the teacher formally reading the KICD-mandated value statement on responsibility and decision-making.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Die rolling probability with independent events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Probability of Independent Events (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner receives an index card. Working individually and silently for 5 minutes, they write their current best answer to the Driving Question: 'If past outcomes of a chance event do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling?' Learners are told this is NOT a test — it is a snapshot of their thinking after eight lessons. They then share with a partner (Think-Pair) for 2 minutes before the teacher takes 3 cold-call shares. Cards are kept for comparison with Lesson 1 predictions.	Display the anchoring phenomenon image on the board: a split-screen of a Gor Mahia FC celebration photo alongside a photo of a matatu in Nairobi with a betting advertisement. Say verbatim: 'Eight lessons ago, this image puzzled us. Today you are going to prove — with mathematics — why the betting advertisement is misleading.' Distribute index cards. Say: 'Write your best answer to our Driving Question. Use as many probability terms as you can. You have 5 minutes — go.' Start a visible 5-minute timer. Circulate silently; do not intervene. After 5 minutes, say: 'Turn to your partner. Compare your answers. Where do you agree? Where is one of you more precise? 2 minutes.' After 2 minutes, cold-call exactly 3 learners using a name-stick draw. For each, ask: 'What ONE mathematical idea is most important in your answer?' Allow 10 seconds of WAIT TIME after each response before probing. Write key terms from responses on the board in three columns: EQUALLY LIKELY RANGE MUTUALLY EXCLUSIVE INDEPENDENT. Say: 'Every term on this board is a tool you have built. Today you will use ALL of them together.'	Individual Prediction Writing (Retrieval Practice): By writing a complete answer before group work, learners activate and consolidate prior knowledge across all eight lessons. The Think-Pair step surfaces vocabulary gaps before public sharing. The teacher's four-column board organiser makes the conceptual architecture of the sub-strand visible, priming learners for the synthesis task ahead.	Observable indicator: Learners' index cards use at least two probability terms correctly in a sentence connecting to the phenomenon. The teacher scans 5–6 cards during circulation and notes whether learners are linking independence to the gambler's fallacy. Cards where learners only describe equally likely events without mentioning independence signal that Phase 3 (Explain) must explicitly bridge this gap.
Observe Phase  VIDEO: General	Groups of 4 receive an 'Evidence Pack' — a single A3 sheet with four worked scenarios, each requiring computation and	Distribute Evidence Packs face-down. Say: 'Do not flip until I say go.' Briefly read each scenario title aloud. Then say: 'These four	Scenario-Based Evidence Analysis: Each scenario is deliberately ordered to mirror the sub-strand's conceptual	Observable indicator: In Scenario 4, groups correctly compute $0.6 \times 0.6 = 0.216$ AND write a sentence explaining that even a

multiplication rule example: independent events Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Probability of Independent Events Practice Source: CK-12 Search ARES for similar readings	interpretation. Scenario 1 (Equally Likely Events): A fair coin is tossed — what is $P(\text{Heads})$? If it has landed on Heads 5 times in a row, what is $P(\text{Heads})$ on the 6th toss? Explain. Scenario 2 (Range of Probability): A Kisumu street vendor claims his lucky number will definitely appear on a 6-faced die. Write the probability of this and any event that has probability 0. Where does every real probability sit? Scenario 3 (Mutually Exclusive Events): In a single Gor Mahia vs AFC Leopards match, what is $P(\text{Gor Mahia wins AND AFC Leopards wins the same match})$? Identify this as mutually exclusive. Scenario 4 (Independent Events): Gor Mahia plays three consecutive matches. Assume each match is independent and $P(\text{Gor Mahia wins}) = 0.6$. Calculate $P(\text{wins all three})$. What does this tell a bettor? Groups work for 12 minutes, recording calculations and written explanations on the Evidence Pack.	scenarios cover everything we have learned. Your job is not just to calculate — it is to explain what each answer means for the person in the scenario. Flip and begin.' Set a visible 12-minute timer. Circulate using a 3-stop rotation: (1) at a group computing Scenario 4, ask: 'What does $P = 0.216$ mean for a person betting on all three matches?' Allow 10 seconds WAIT TIME. (2) At a group on Scenario 1, ask: 'Does the coin remember what happened before?' — pause 10 seconds — then: 'Show me in your working.' (3) At a group on Scenario 3, ask: 'Can both teams win the same match? What word describes that?' After 12 minutes, say: 'Complete your calculations. Underline the ONE sentence in each scenario that best explains the mathematics to a non-mathematician.' Cold-call 2 groups to share their underlined sentence for Scenario 4 before moving on.	progression (equally likely → range → mutually exclusive → independent), allowing learners to see how the concepts stack into a complete explanatory framework. The instruction to 'explain to a non-mathematician' shifts learners from procedural computation to genuine sensemaking, directly preparing the Explain Phase mini-poster.	60% chance per match gives only a ~22% chance of winning all three, illustrating why repeated betting is a losing strategy. Groups that calculate correctly but write no explanation need targeted prompting in Phase 3. Groups that multiply incorrectly (e.g., add probabilities) require immediate re-teaching of the multiplication rule for independent events.
Explain Phase VIDEO: Die rolling probability with independent events - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos HTML: Independent, Conditional, and Mutually Exclusive Events (Flexbook)	Each group uses a large sheet of paper (A2 or flip-chart) to construct a mini-poster that answers the Driving Question in full. The poster must include: (a) a title restating the Driving Question in their own words; (b) one worked example for each of the four concepts, all using Kenyan contexts from the Evidence Pack or their own; (c) a clear written explanation of the gambler's fallacy using the word 'independent'; (d) a mathematical argument for why gambling is a losing proposition (using the multiplication rule for independent events); (e) a probability line (0 to 1) with at least three events from	Before groups begin, display the poster requirements on the board and say verbatim: 'Your poster must make a mathematician AND a Standard 8 pupil both understand why the matatu betting ad is misleading. Use numbers, words, and a probability line.' Set a 15-minute timer. During construction, circulate and use targeted probing at 3 groups: (a) 'Where is the word INDEPENDENT on your poster — and why does it matter?' (b) 'Your probability line — where would you place $P(\text{Gor Mahia wins every match for 10 games in a row})$? Calculate it.' (c) 'What is the	Gallery Walk with Structured Feedback Cards: The gallery-walk format externalises group thinking for peer critique, which is a high-leverage sensemaking strategy because learners must evaluate mathematical precision in others' arguments — a skill that deepens their own understanding. The 'revision in a different colour' step makes the improvement process visible and models scientific revision of explanations. The teacher's anonymous reading of three gambler's fallacy explanations creates a whole-class norm-setting moment that closes the conceptual arc of the sub-	Observable indicator: Mini-posters include a correct, written statement of the form 'Because each match is an independent event, $P(\text{win}) = 0.6$ regardless of previous results' AND a correct multiplication calculation showing how probability decreases across consecutive independent bets. Feedback cards that only reference visual design rather than mathematical content indicate learners are not yet evaluating conceptual precision — the teacher should redirect by asking: 'Point to the calculation. Is it correct? Is the explanation complete?'

Source: CK-12  Search ARES for similar readings	the phenomenon placed on it. Groups have 15 minutes to build the poster, then conduct a Gallery Walk: posters are displayed around the room, one group member stays as 'explainer' while the other three walk. Walkers use a structured feedback card with two prompts: 'The strongest mathematical argument on this poster is ____' and 'One thing that could be more precise is ____'. After 8 minutes of walking, groups reunite, read feedback, and make one revision to their poster in a different colour.	difference between mutually exclusive and independent? Can you show both on one poster?' After gallery walk, bring the class together. Say: 'I am going to read three posters' gambler's fallacy explanations aloud without naming the group. Vote with thumbs — which is most mathematically precise?' Read 3 excerpts. After each, allow 10 seconds WAIT TIME before thumbs. Announce the most precise and explain why, using KICD language: 'Independent events have no influence on each other's probability — that is the mathematical truth the betting industry hopes you forget.'	strand.	
Driving Question Board (DQB) Creation  VIDEO: Compound probability of independent events - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Independent probability Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The full DQB (constructed and added to across all 8 lessons) is displayed. Each group receives three sticky dots: green (answered fully), yellow (partially answered), red (still wondering). Working as a class, learners go question by question. For each question, the teacher reads it aloud and asks: 'Has our learning answered this? Who can point to the evidence — on a poster, in their notes, or in today's Evidence Pack?' Learners place their dots. Questions that receive all green dots are ceremonially 'closed' by the teacher drawing a tick and writing the key mathematical answer beside them in one sentence. Questions receiving red dots are acknowledged as productive wonderings that point beyond Grade 10 (e.g., conditional probability, expected value). The Driving Question itself is read aloud by the whole class in unison, then the teacher cold-calls 3 individual learners to each give	Stand at the DQB. Say: 'This board has been our scientific record for 8 lessons. Every question on it came from YOUR curiosity about Gor Mahia and the matatu betting ads. Let's see how far our mathematics has taken us.' Read the first question. Pause 10 seconds. Ask: 'Green, yellow, or red — dot it.' After all questions are dotted, focus on any red-dot questions. Say: 'These are not failures — they are invitations. The mathematics you will study in Grade 11 and 12 will answer them.' Write beside each red-dot question: 'See: conditional probability / expected value — Grade 11.' For the Driving Question closure, say: 'Everyone read the Driving Question with me — together.' Lead the unison read. Then say: 'I need three voices to build our final answer. [Name 1], give me the first sentence — about equally likely events and independence.' WAIT 10 seconds. '[Name 2], add a sentence about	DQB Dot Voting and Composite Answer Construction: The dot-voting protocol makes the class's collective epistemic progress visible — learners can literally see how many of their original questions are now answered with mathematics. The composite three-sentence final answer, built from three cold-called voices, models the scientific practice of constructing explanations from evidence and is deeply connected to NGSS SEP 6 (Constructing Explanations). Writing it on the board and in notes creates a durable record of consolidated understanding.	Observable indicator: The three cold-called learners each produce a grammatically complete, mathematically accurate sentence that uses at least one probability term correctly and references the phenomenon. If a called learner produces a vague sentence (e.g., 'probability means chance'), the teacher probes: 'Can you add a number or a formula to make that more precise?' and allows 10 additional seconds of WAIT TIME before scaffolding.

	one sentence of the answer, building a class composite final answer that is written on the board.	the range of probability and what it tells a better.' WAIT 10 seconds. '[Name 3], close it — why is gambling mathematically a losing proposition?' Write the three-sentence composite answer on the board. Say: 'This is our class's final answer. Copy it into your notes.'		
Model Building Phase  VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Probability of Independent Events (Flexbook) Source: CK-12  Search ARES for similar readings	Part A — Model Comparison (8 min): Learners retrieve their Lesson 1 initial model (a drawing/diagram of what they thought governed football match outcomes). They draw a revised Final Model beside it, incorporating all four probability concepts, the independence of each match, the probability line, and the gambler's fallacy label. They annotate with arrows showing what changed and why. Part B — Written Assessment (20 min): Learners complete a short individual written assessment (5 questions) covering all SLOs: (1) Compute P(equally likely event) in a dice scenario; (2) Place three events on a probability line and justify; (3) Identify two mutually exclusive events from a Kenyan football context; (4) Calculate P(independent events) across two trials using a coin and a Gor Mahia match; (5) Write 3 sentences explaining why the gambler's fallacy is mathematically false. Part C — Personal Reflection (5 min): On the back of the assessment paper, learners complete: 'One real-life decision I will make differently because of what I now know about probability is ___, because mathematically ___.'	Distribute Lesson 1 model sheets (collected earlier in the unit). Say: 'Look at what you thought 8 lessons ago. Now draw your final model beside it. Use colour — red for what you now know was wrong, green for what you now know is mathematically true.' Set 8-minute timer. Circulate, noting 3 learners whose models show the most dramatic conceptual growth — affirm verbally and privately: 'This revision shows real mathematical thinking.' After model comparison, collect models and distribute assessment papers. Say clearly: 'This is individual work. No group discussion. You have 20 minutes. Use every probability term we have learned. Show all working.' Set timer. Circulate silently for the first 5 minutes, then stop and stand at the front. For Question 5 (gambler's fallacy explanation), after 10 minutes say: 'If you are stuck on Question 5, look at the composite answer on the board — but write it in YOUR own words.' After assessments are collected, say: 'Flip your paper. One minute of silence. Think of one real decision — buying a scratch-card, betting on a match, playing a game of chance. Now write your reflection. This is for you.' Allow 5 minutes. Close the lesson by reading verbatim from KICD: 'The learner makes correct decisions on life issues that involve gambling	Model Comparison (Initial vs. Final): Placing the Lesson 1 model and Final Model side by side makes the learner's conceptual growth concrete and visible — a metacognitive strategy that research consistently links to deep learning. The colour-coded annotation (red = corrected misconception, green = new understanding) makes the revision process explicit rather than implicit. The personal reflection connects mathematical knowledge to agency and decision-making, fulfilling the KICD PCI of Life Skills (Decision Making) and the value of Responsibility.	Observable indicators for Written Assessment: (Q1) Correct probability fraction with sample space shown. (Q2) Three events correctly placed with $P = 0$, $0 < P < 1$, and $P = 1$ each represented. (Q3) Two events named that genuinely cannot occur simultaneously (e.g., 'Gor Mahia wins AND loses the same match'). (Q4) Correct multiplication of two independent probabilities (e.g., $0.5 \times 0.6 = 0.3$) with working shown. (Q5) Three sentences that include the word 'independent', a statement that past outcomes do not change future probability, and a reference to the phenomenon. Learners who score below 3/5 are flagged for a 15-minute catch-up session using the Evidence Pack scenarios as remediation tools.

		using the knowledge of probability.' Say: 'That is what you have done today. That is mathematics in service of your life.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your notes:

1. **CONCEPTUAL GROWTH:** Compare three learners' Lesson 1 models with their Final Models. What is the most common misconception that PERSISTED despite eight lessons? (Common residual error: learners may still confuse 'independent' with 'mutually exclusive' — if so, plan a 10-minute re-teach opener for the next strand using a Kenyan context, e.g., drawing a card from a deck vs. tossing a coin simultaneously.)
2. **DQB QUALITY:** Were there questions on the DQB that received red dots (unanswered)? List them. Do they suggest gaps in your instructional sequencing, or are they genuinely beyond Grade 10 scope? Plan how to explicitly bridge these to Grade 11 content in your end-of-unit notes to learners.
3. **ENGAGEMENT WITH GAMBLING PCI:** Did the personal reflection task (Part C) reveal any learners whose family context involves gambling? If so, consult the school counsellor. The mathematics of probability is a powerful protective tool — ensure the framing remained empowering, not shaming.
4. **ASSESSMENT DATA:** Tally scores on Q4 (independent events multiplication) and Q5 (gambler's fallacy explanation). If more than 30% of learners score 0 on Q5, the verbal/written explanation skill was under-practised across the unit — plan more 'explain to a non-mathematician' tasks in future sub-strand sequences.
5. **GALLERY WALK FEEDBACK CARDS:** Review the structured feedback cards collected from the gallery walk. Did learners evaluate mathematical precision or only surface features (neatness, colour)? If the latter, model mathematical peer feedback explicitly at the start of the next unit.
6. **KICD VALUES CHECK:** Did the closing statement — reading the KICD value of Responsibility aloud — feel authentic in your classroom context? If learners responded with recognition (nodding, quiet), it landed. If it felt formal, consider asking a learner to read it next time to increase ownership.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners worked through four Kenyan-context probability scenarios (coin toss independence, Kisumu dice range, mutually exclusive football match outcomes, and a three-match independent-event multiplication), recorded calculations and written explanations on an Evidence Pack, and conducted a gallery walk critiquing peer mini-posters using structured feedback cards.
What did I learn?	Each football match (and every chance event studied) is an independent event — its probability is fixed and cannot be changed by past outcomes. The gambler's fallacy is mathematically false because $P(\text{event})$ for an independent trial does not depend on previous results. Gambling is a losing proposition because the multiplication rule for independent events shows that $P(\text{winning multiple consecutive bets})$ decreases sharply with each additional trial. All probability values lie in the range $0 \leq P(E) \leq 1$, and mutually exclusive events (both teams winning the same match) have a combined probability of zero.
How does this explain the phenomenon?	Using all four probability concepts together — equally likely events, range of probability, mutually exclusive events, and independent events — learners constructed a complete written and visual explanation of why the Gor Mahia winning-streak debate and the Nairobi matatu betting advertisements are both examples of the gambler's fallacy. The composite class answer, the mini-posters, the DQB closure, and the written assessment all demonstrate that mathematics — not luck, streaks, or intuition — governs the odds, and that this knowledge is a practical tool for responsible decision-making in real life.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.